Lecture 09

Carbon Tariffs

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AEM 4510

Roadmap

1. What are carbon tariffs?
2. Basic trade theory: supply, demand, and tariffs
3. Adding pollution externalities to trade
4. Unilateral carbon taxes and leakage
5. Carbon tariffs to prevent leakage
6. Carbon tariffs alone
7. World carbon taxes
8. Different pollution intensities across countries

What are carbon tariffs?

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Carbon tariffs aim to "level the playing field" and reduce leakage

Carbon tariffs in practice

EU Carbon Border Adjustment Mechanism (CBAM):

- Reporting began in 2023, the definitive regime began in January 2026
- Covers cement, iron/steel, aluminum, fertilizers, electricity, hydrogen
- Importers must purchase certificates matching EU ETS price

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UK CBAM: starts in January 2027

US proposals: the CCA and FPFA remain in committee, neither is law

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- The benchmark starts at the US industry's average emissions intensity
- It tightens over time, encouraging cleaner production
- The charge starts at 60 dollars per metric ton of CO_2e above the benchmark
- Revenue supports cleaner industrial production

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Sponsors emphasize US manufacturing competitiveness and competition with China

This connects to the tariff alone case we will study later

Comparing the two US proposals



Watch video

Understanding border carbon adjustments



Watch video

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3. What complications arise in practice

The key insight: carbon tariffs are equivalent to a **consumption tax** plus a **production subsidy**

Basic trade theory

Tariffs are taxes on imports

A tariff is a **tax on imports**

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A tariff is a **tax on imports**

In this lecture we study the small open country case:

- Small: country i is too small to affect the world price
- Open: the country trades goods with the rest of the world
- Imports are available at the fixed world price P_W, regardless of how much this country imports
- This is what makes leakage especially transparent later on

A simple trade model

Consider a small open economy (country i) that:

- Takes world prices P_W as given
- Can import or export a homogeneous good

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Key curves:

- S: domestic supply
- D: domestic demand
- P_W: world price
  - The social marginal benefit of the good

Domestic suppliers can't charge more than P_W, and won't charge less

Supply and demand refresher

In this market:

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Supply and demand refresher

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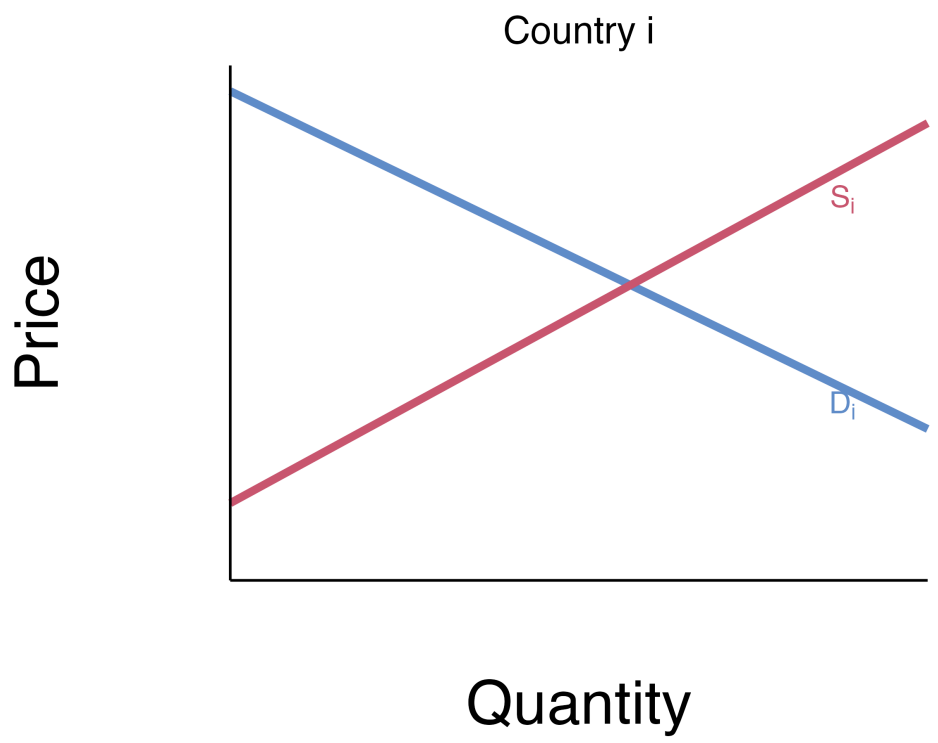
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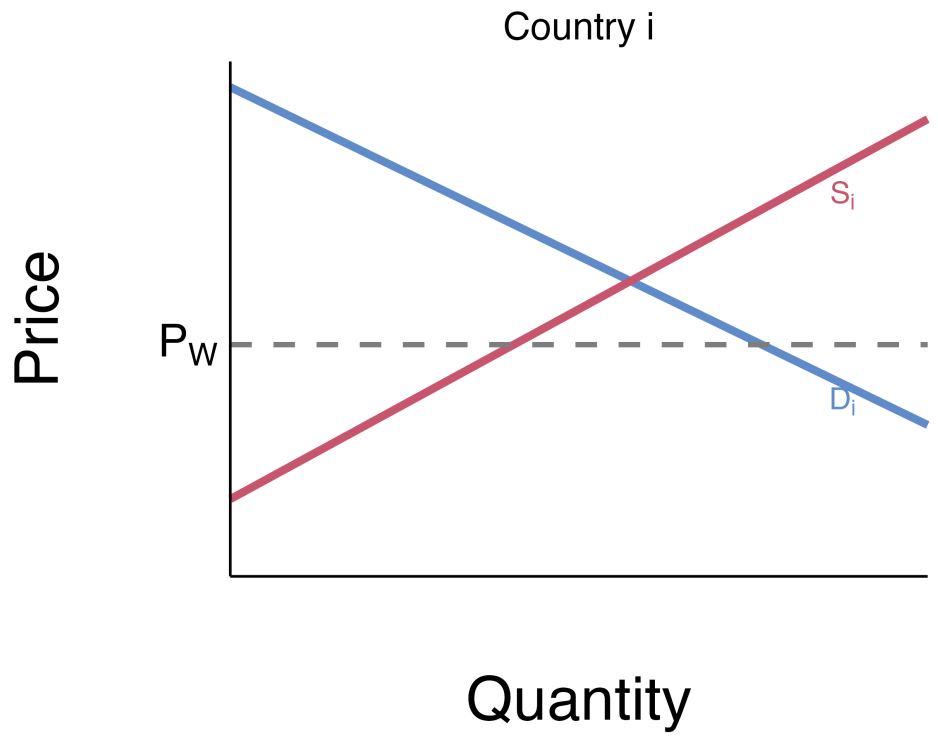
- D is **marginal benefit** to domestic consumers
- S is **marginal cost** to domestic producers
- If P_W is below the autarky (no trade) price, country i imports the good
- The gap between domestic demand and domestic supply is imports

Free trade benchmark



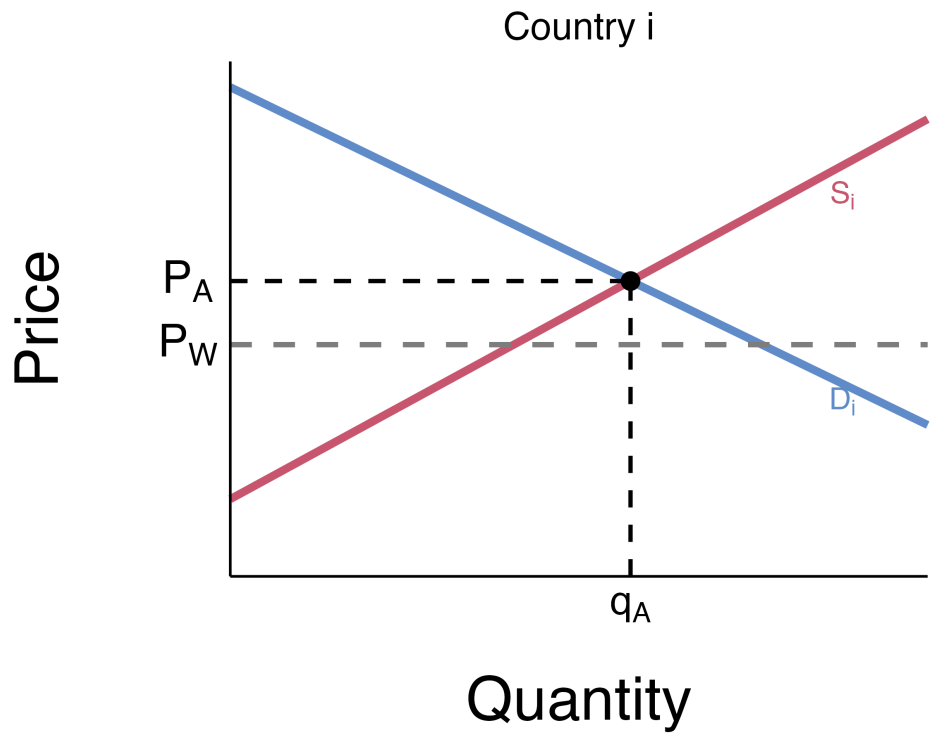
Start with domestic supply S and demand D

Free trade benchmark



Free trade makes imports available at the world price P_W

Free trade benchmark

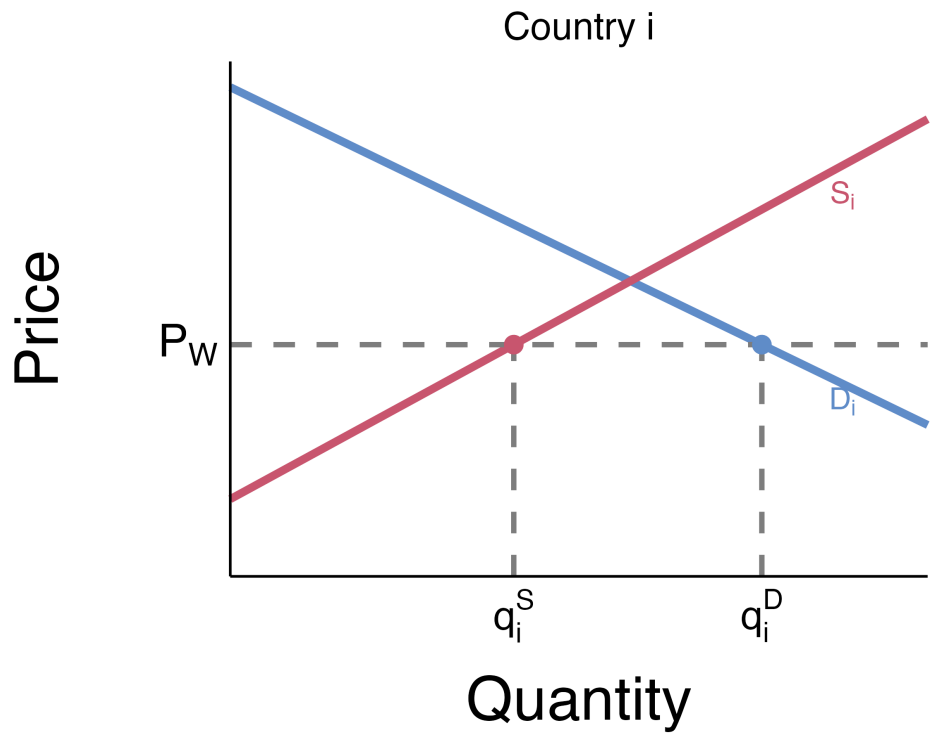


Without trade (autarky), the price is P_A

Domestic supply equals domestic demand: there is no surplus or shortage

The world price P_W is lower, so the country imports with free trade

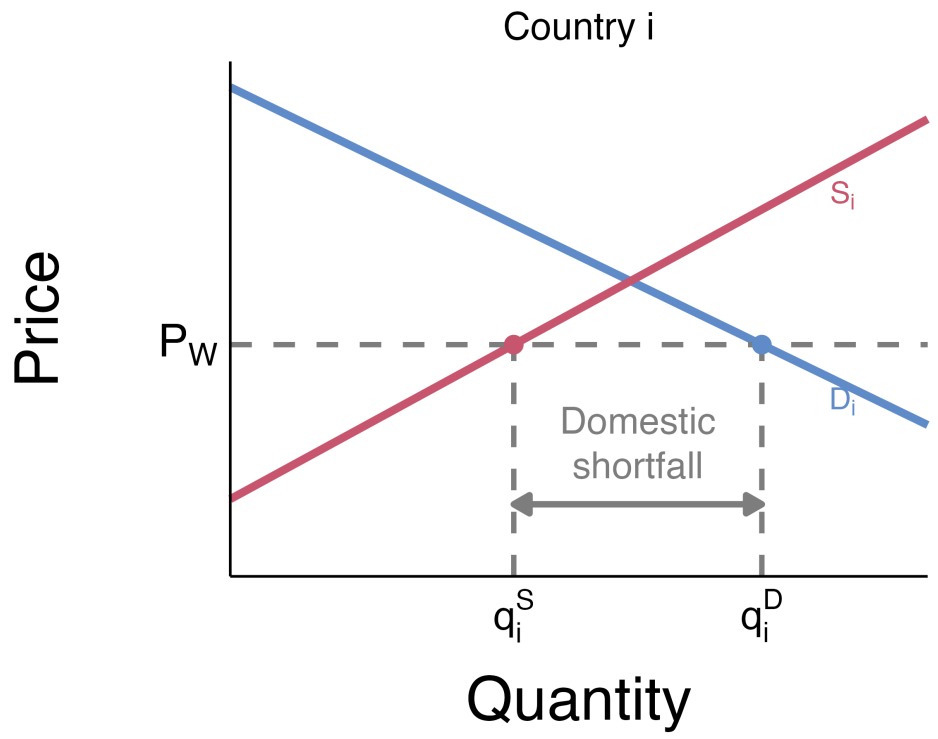
Free trade benchmark



At P_W:

- Consumers want to buy q_i^D
- Producers want to sell q_i^S

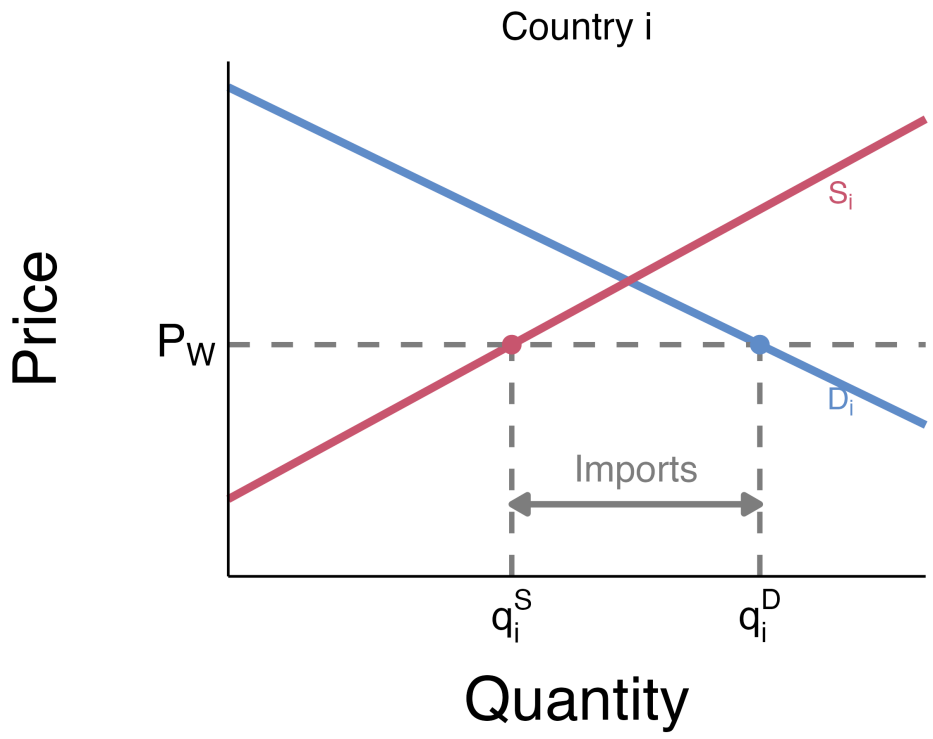
Free trade benchmark



At P_W, consumers want more than domestic producers supply

The domestic shortfall is $q_i^D - q_i^S$

Free trade benchmark

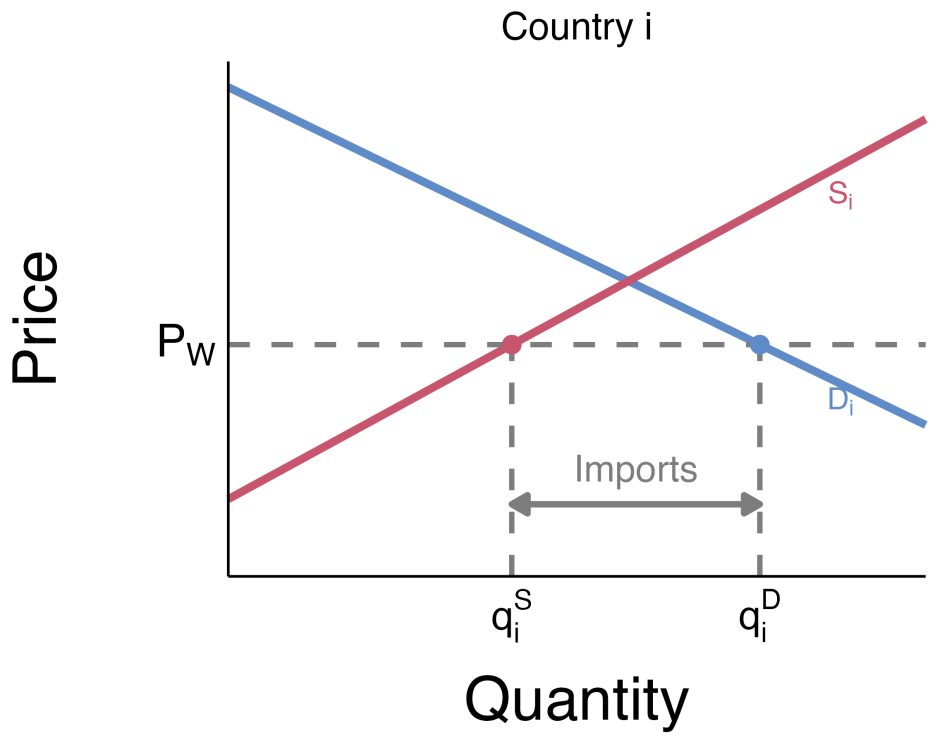


Imports fill the domestic shortfall

$$\textbf{Imports} = q_i^D - q_i^S$$

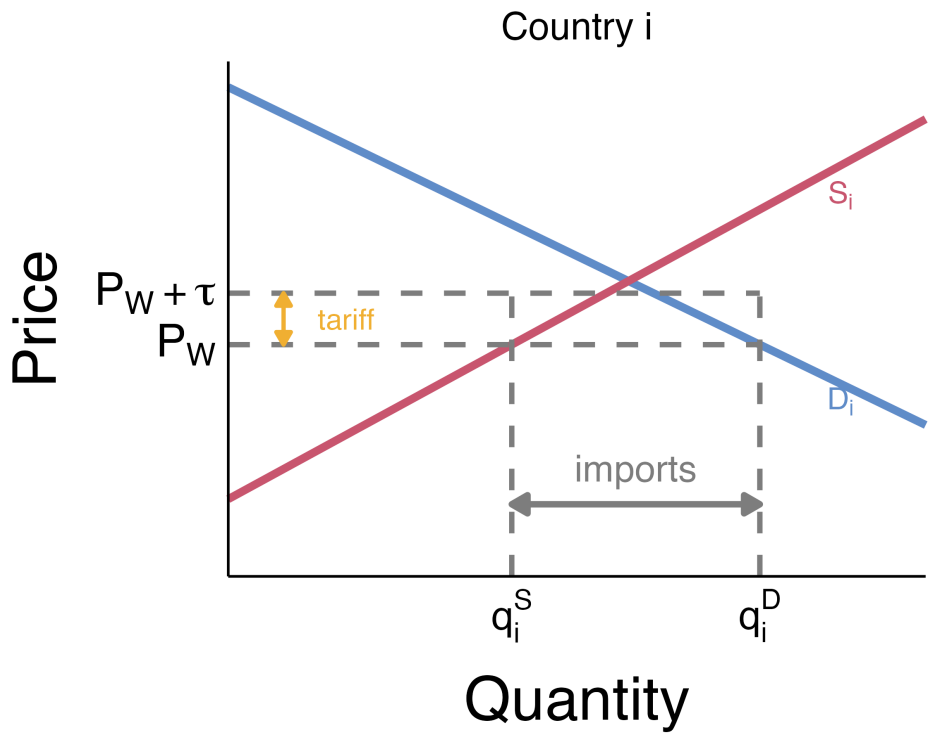
Domestic production plus imports meets domestic demand

The tariff



Suppose we add a tariff of τ dollars per unit on top of the imported good's price P_W

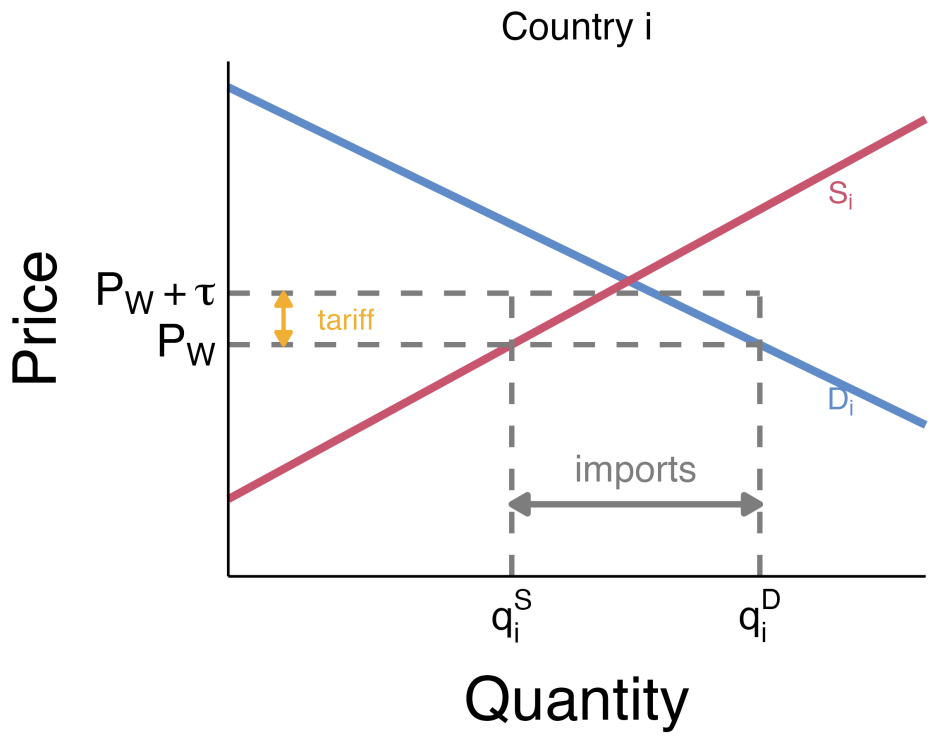
The tariff



It now costs $P_W + \tau$ to buy the imported good

What do domestic producers do?

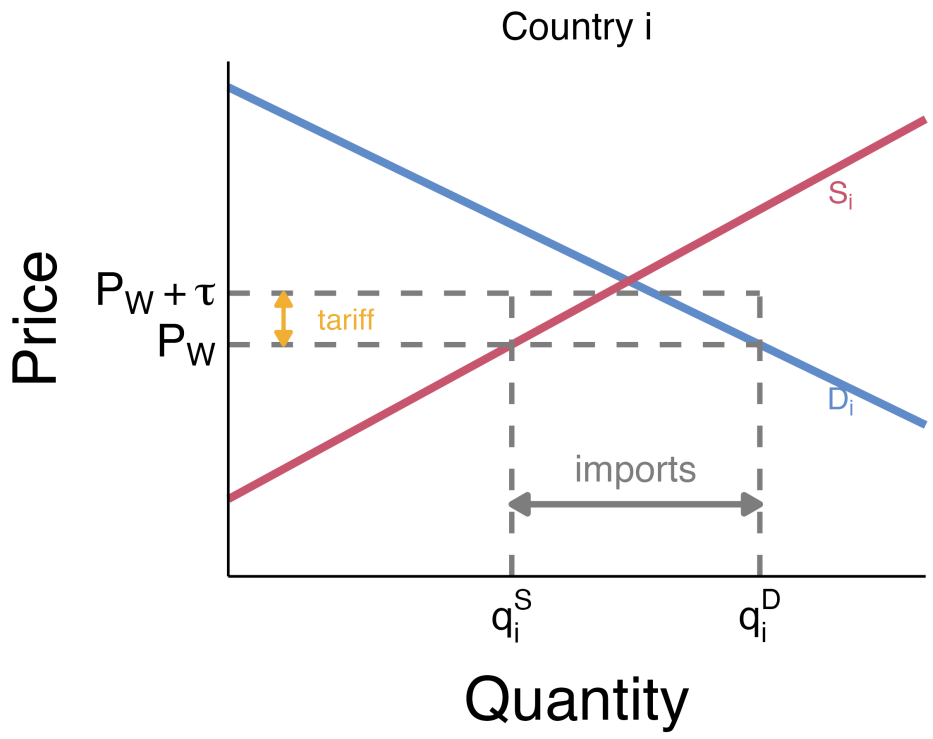
The tariff



Domestic producers can now raise their price to $P_W + \tau$

Imports no longer offer consumers a cheaper alternative

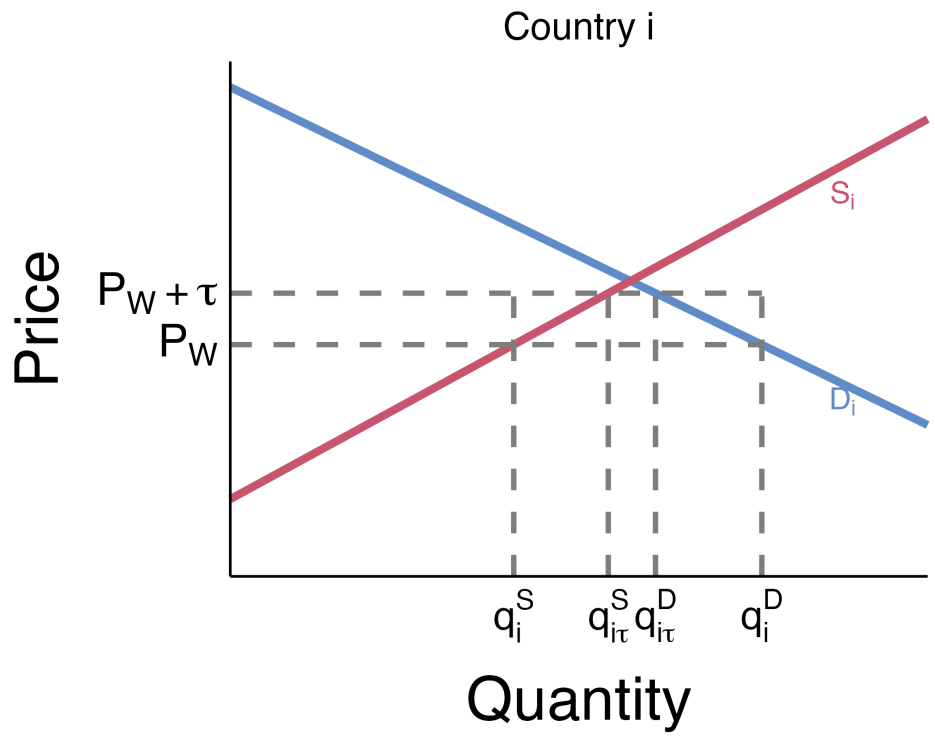
The tariff



Consumers face a price of $P_W + \tau$ no matter who they buy from

The domestic price rises by τ

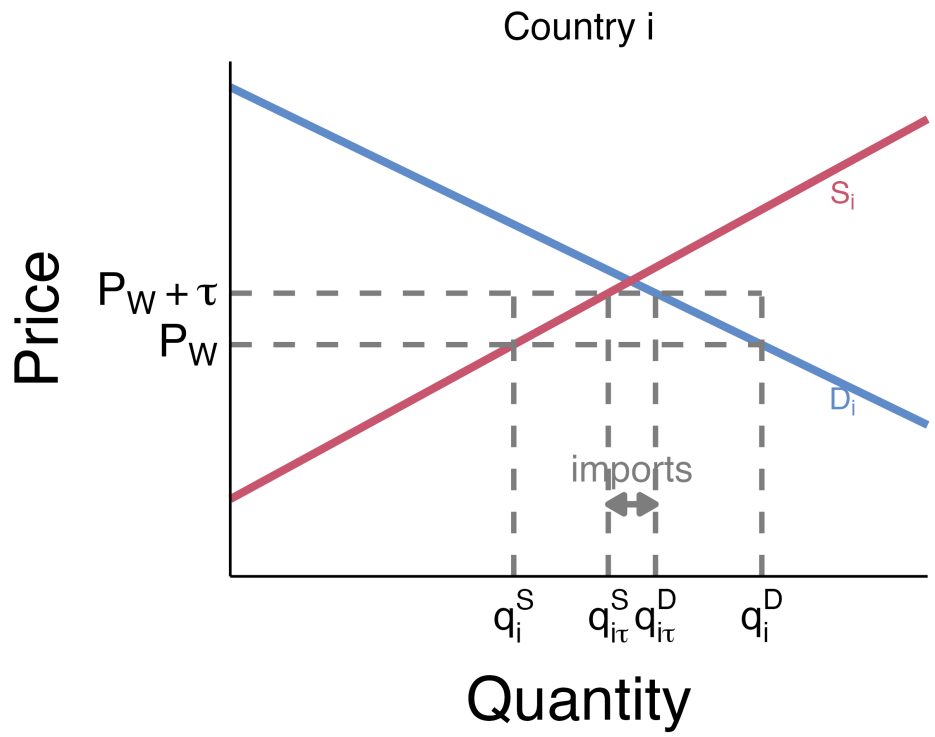
Quantity effects of a tariff



At the higher price:

- Domestic producers sell more
- Consumers buy less

Quantity effects of a tariff

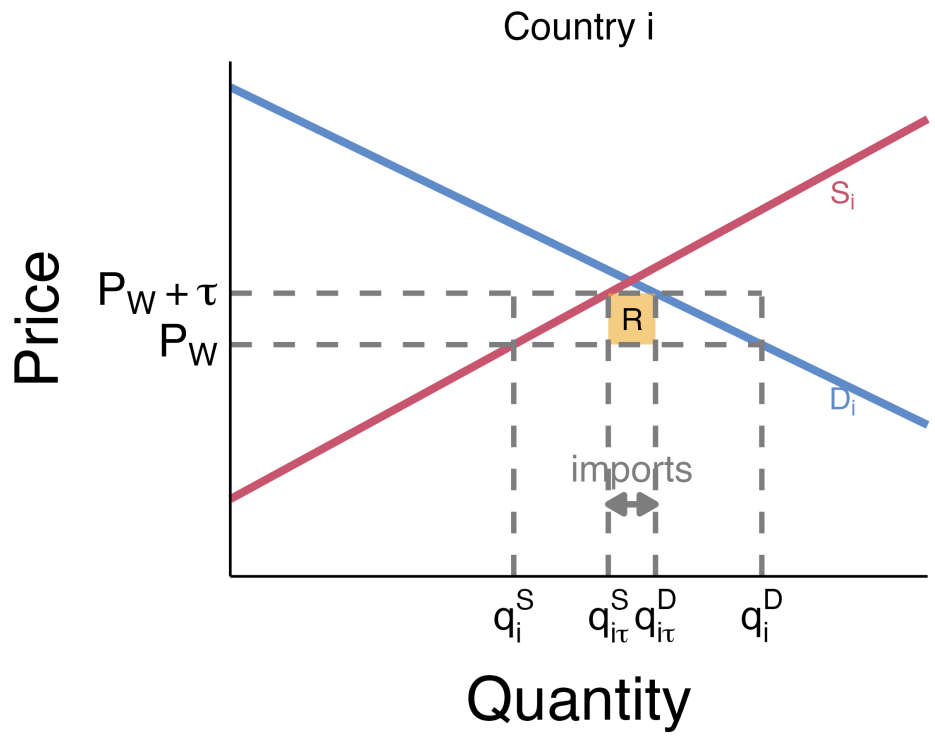


Imports shrink

More domestic production and less consumption both reduce imports

Reducing imports is often the stated purpose of tariffs, as in Trump's focus on trade deficits

Tariff revenue

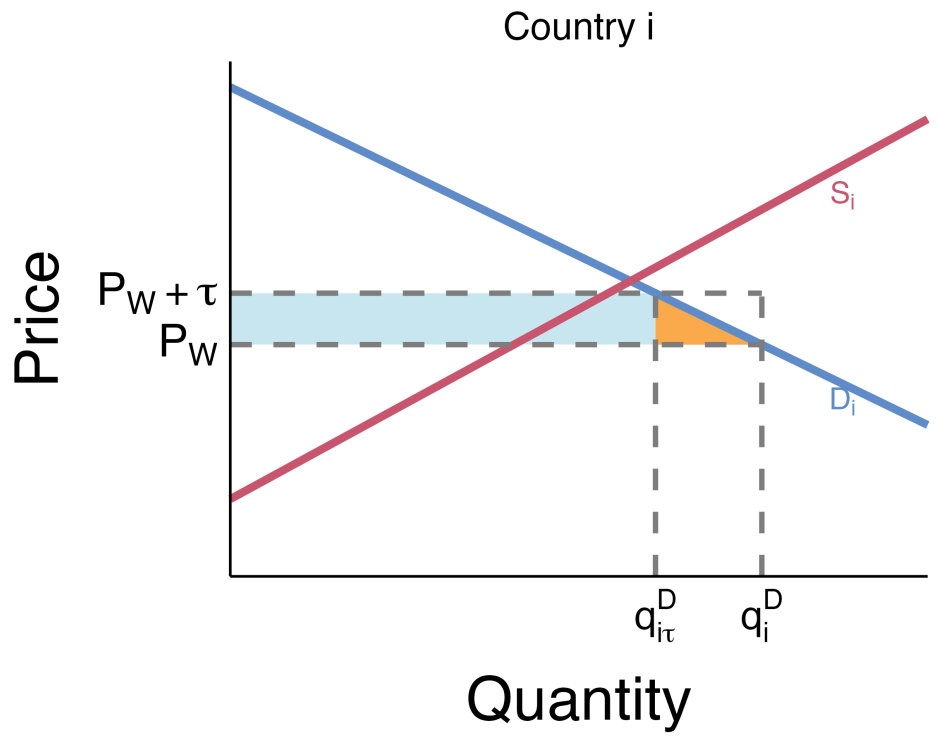


The government collects the tariff on each unit imported

$$R = \tau \times \left(q_{i\tau}^D - q_{i\tau}^S\right)$$

The shaded rectangle is tariff revenue: the tariff per unit times imports

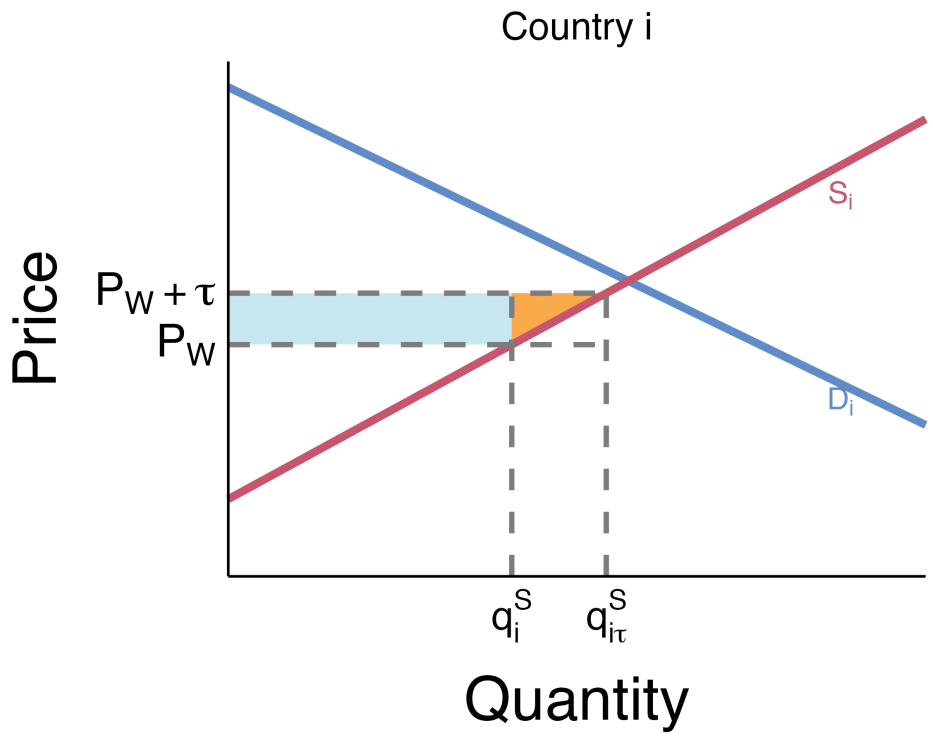
Consumer surplus refresher



Consumer surplus is the area below demand and above price

- Blue rectangle: consumers pay more on the units they still buy
- Orange triangle: consumers stop buying units they still valued above P_W

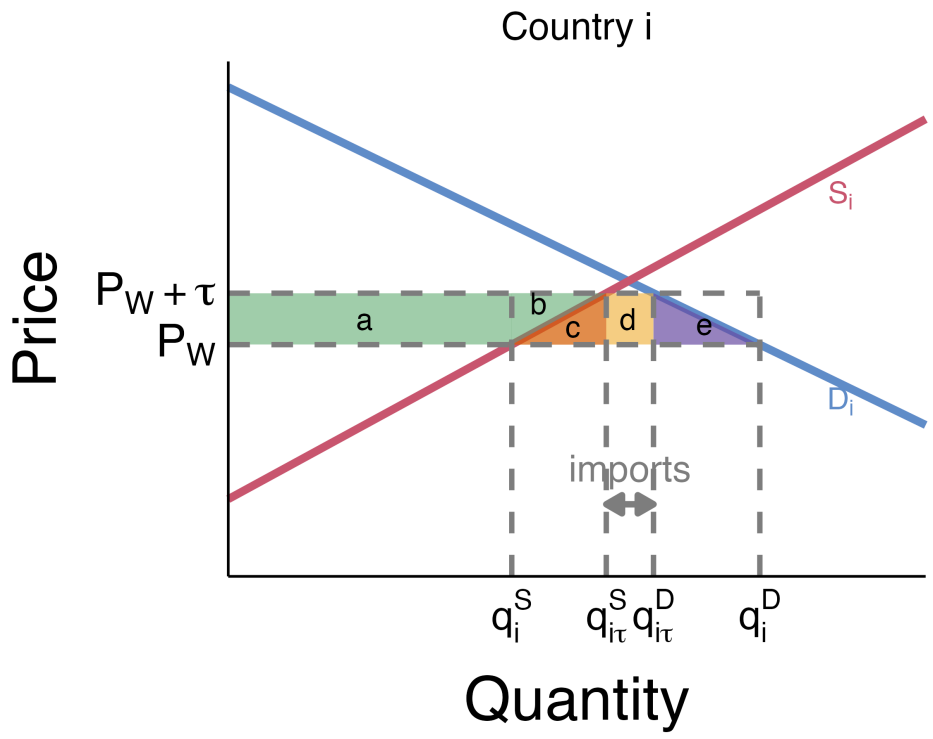
Producer surplus refresher



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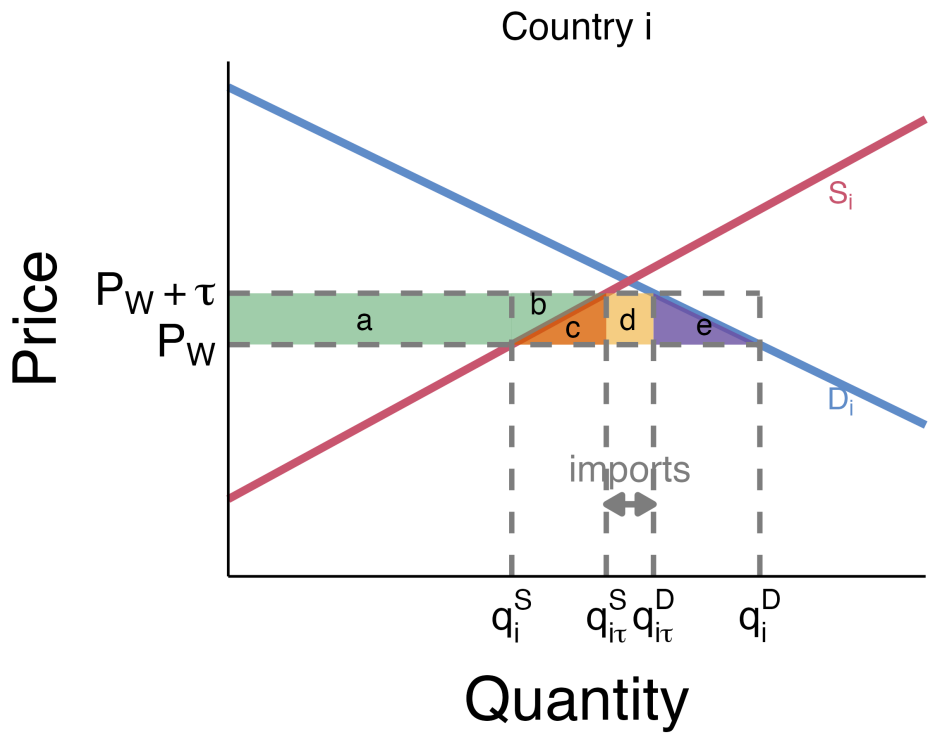
- Blue rectangle: producers receive a higher price on their original output
- Orange triangle: producers gain extra surplus when they expand output to $q_{i\tau}^S$

Consumer loss and redistribution



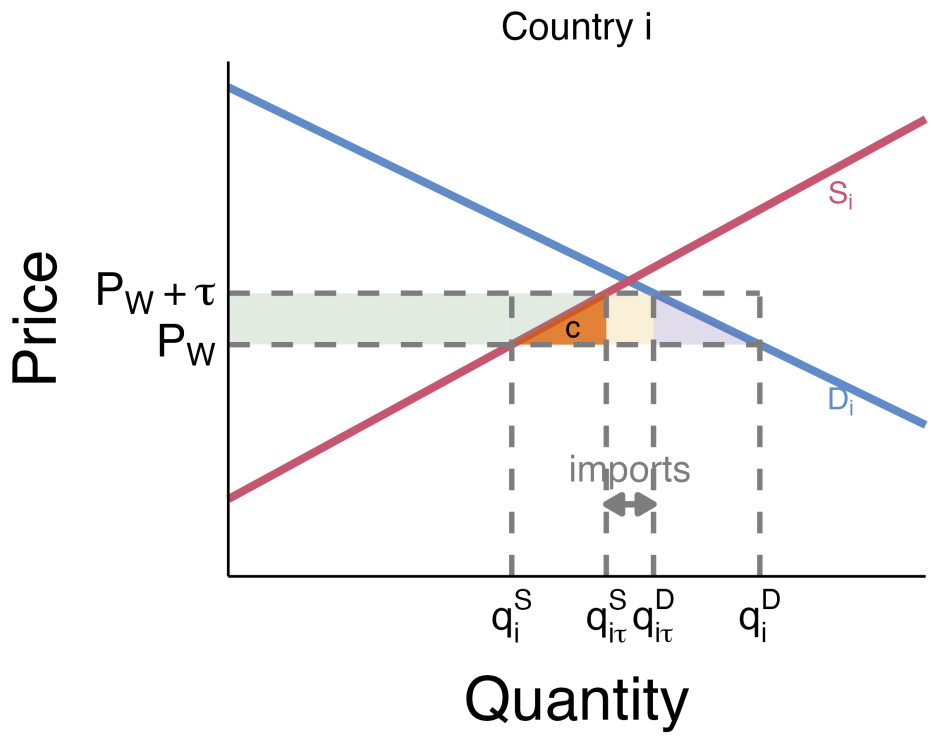
- $-\Delta CS = a + b + c + d + e$
- The whole shaded area is the **consumer surplus loss**
- But $a + b$ reappear as part of **producer surplus gain**
- And d reappears as **tariff revenue**

Tariff welfare accounting



- $\Delta PS = a + b$
- $\Delta CS = -(a + b + c + d + e)$
- $R = d$ is tariff revenue
- $\Delta W_i = \Delta PS + R + \Delta CS$
- So for a **small country**, $\Delta W_i = -c - e$
- The only net welfare losses are the two deadweight loss regions c and e

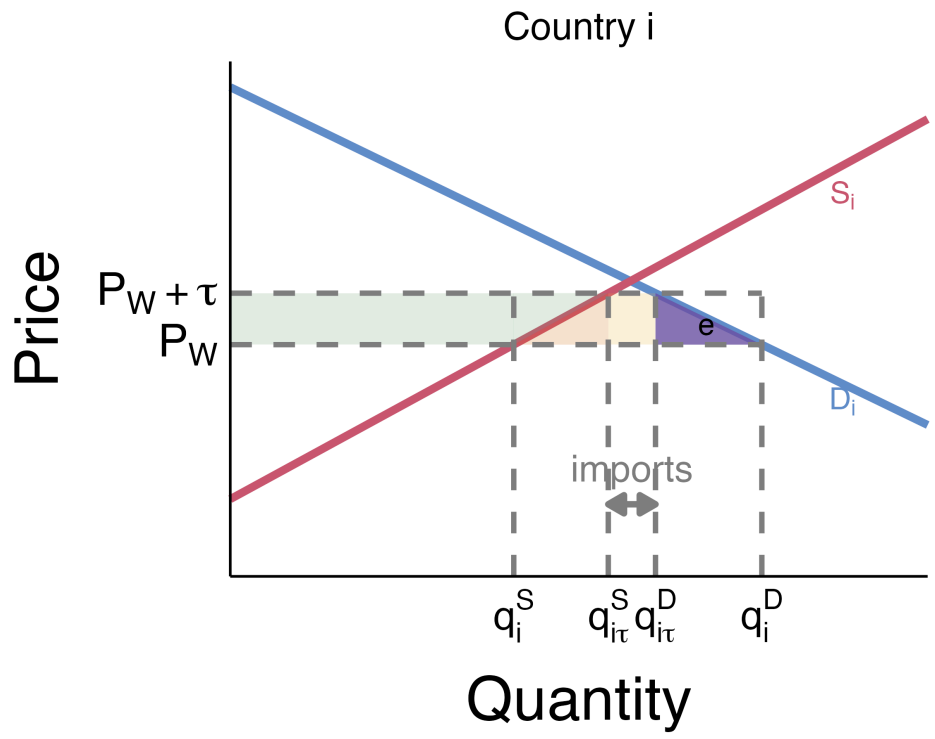
Production distortion



Triangle c: producing in the more expensive country

- The tariff shifts production from cheaper foreign producers to more costly domestic producers
- The extra production cost is deadweight loss

Consumption distortion



Triangle e: consuming too little of the good

- Consumers stop buying units they value more than the world price P_W
- These lost gains from consumption are deadweight loss

Why start with ordinary tariffs?

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So next we add the environmental externality to this standard tariff

Adding pollution

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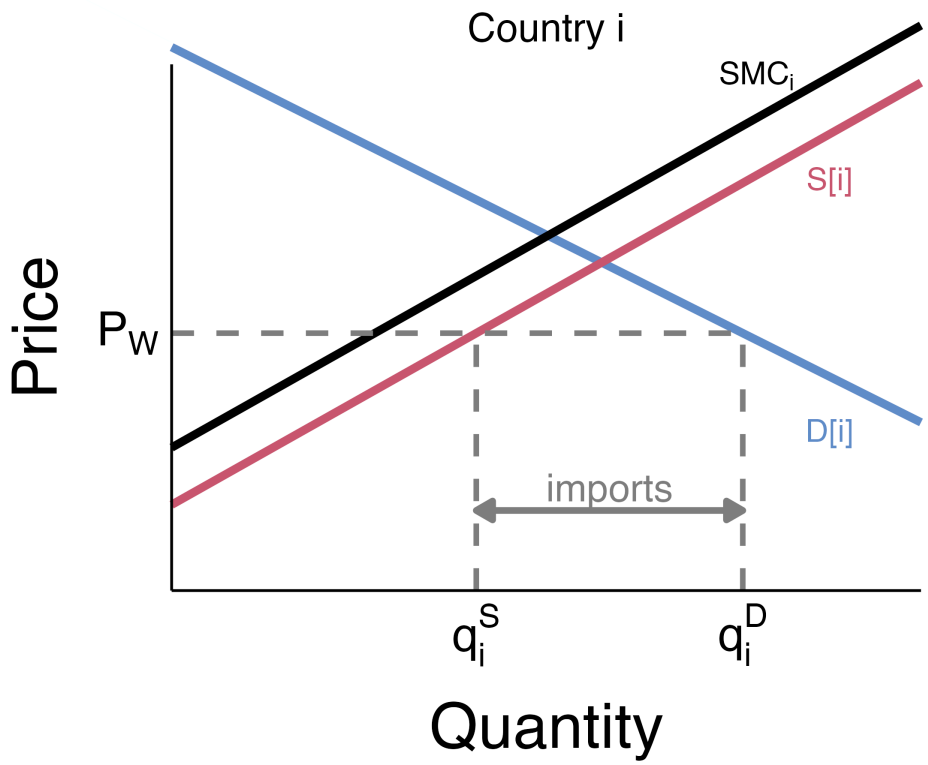
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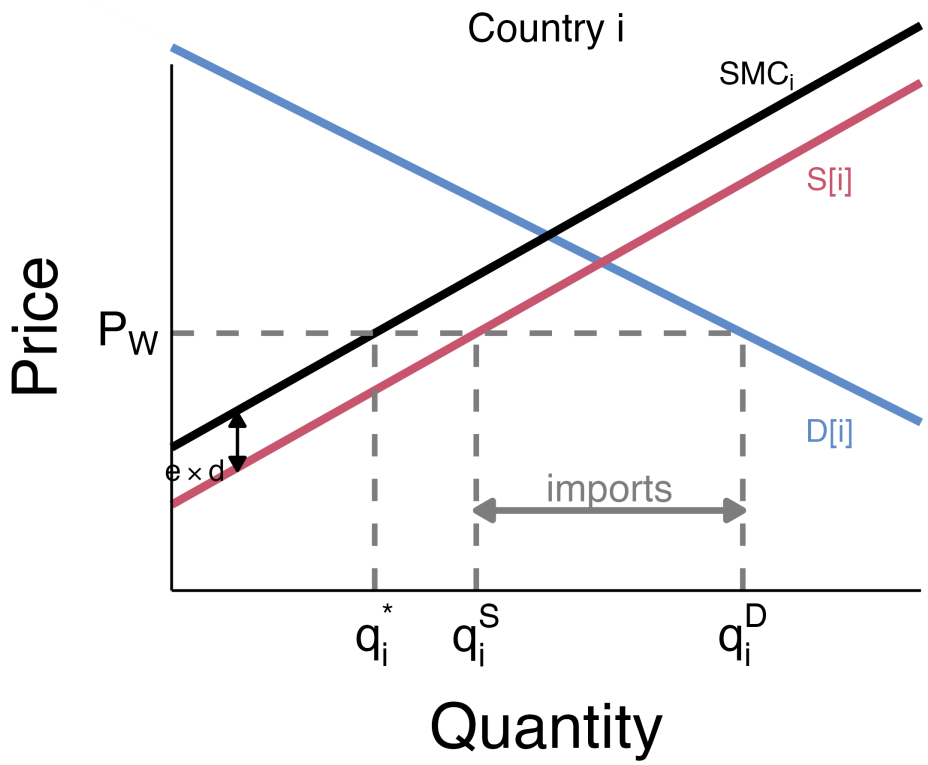
SMC lies above the private supply curve by the external cost $e \times d$

Free trade with pollution



Free trade output is pinned down by domestic demand and supply at the world price

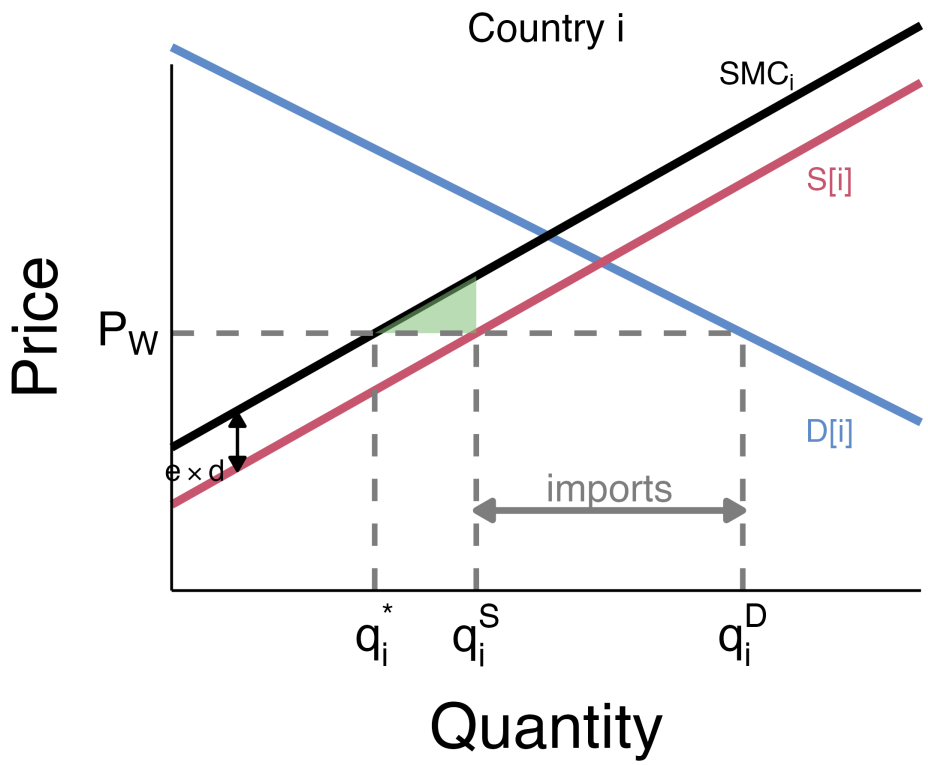
Free trade with pollution



Country i's SMC curve lies above its private supply curve

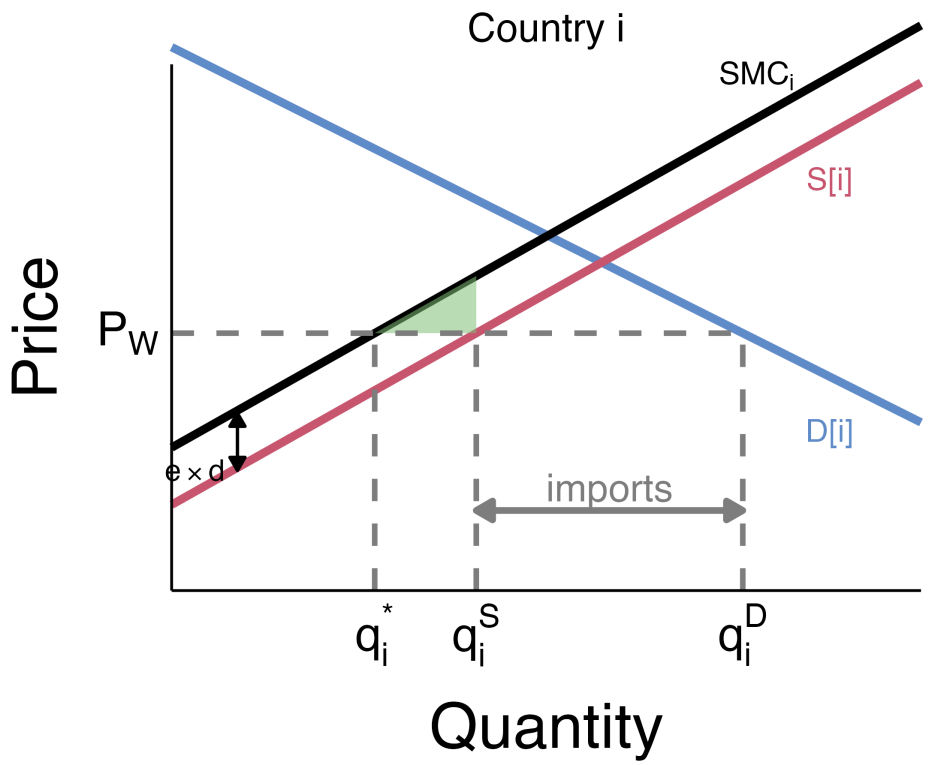
The vertical gap is the external cost: $e \times d$

Free trade with pollution



The green region shows the welfare loss from excessive domestic production, counting global environmental damages and holding foreign production fixed

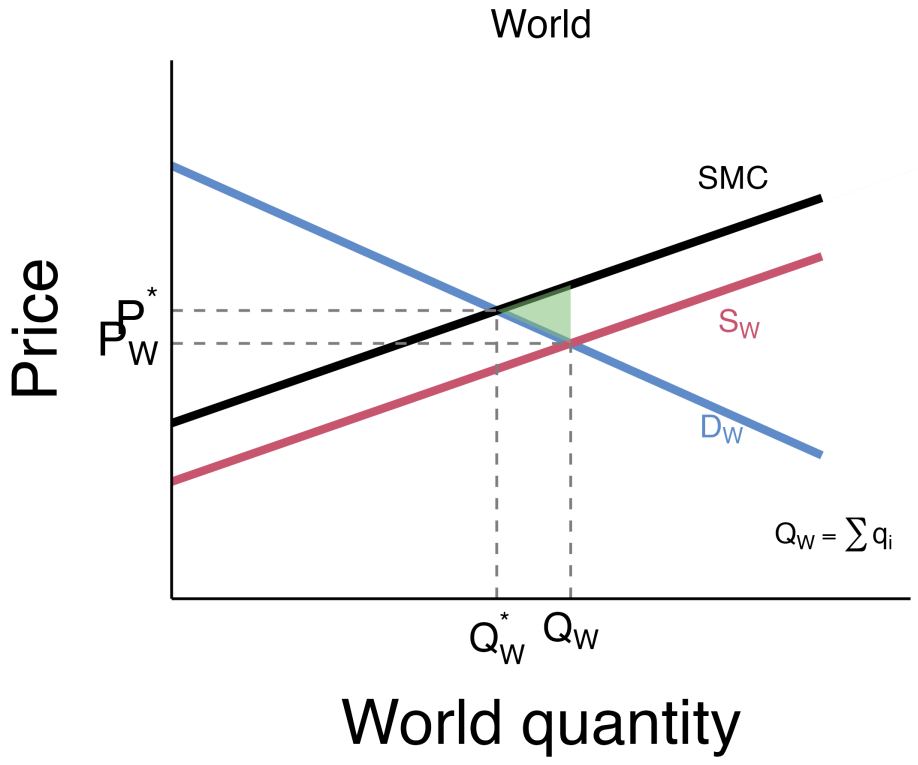
Free trade with pollution



At the world price P_W, marginal benefit is pinned down by world demand, while marginal social cost is given by SMC

That triangle is the set of units for which marginal benefit exceeds private supply cost but is below marginal social cost

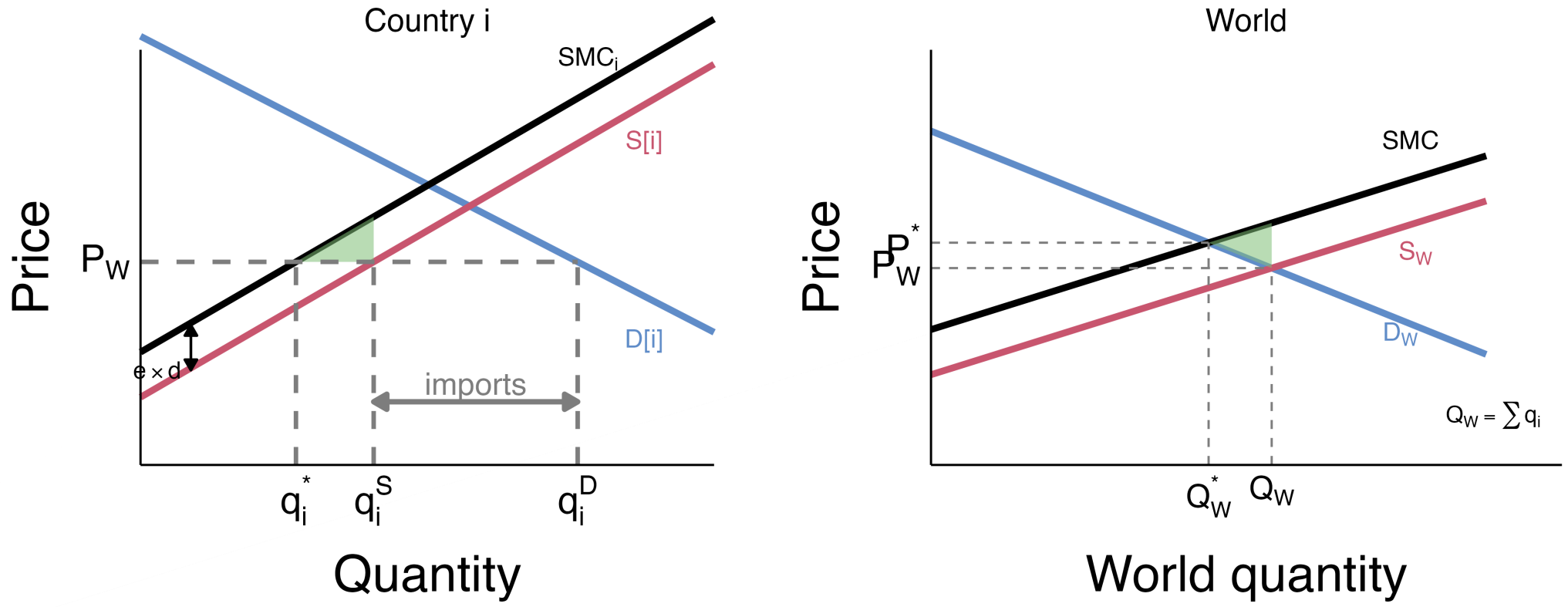
Global perspective



From a global perspective:

- Free trade output is where $D = S$
- Efficient output is where $D = SMC$
- The green region is the global welfare gain from cutting output from Q_W to Q_W^*

Country and global perspectives



Global DWL on the right is the sum of each country's DWL on the left

The externality problem

With a global pollutant like CO_2:

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But what if only **one country** acts?

Unilateral carbon tax

Country i imposes a carbon tax

Suppose country i imposes a carbon tax $t = e \times d$ on domestic production

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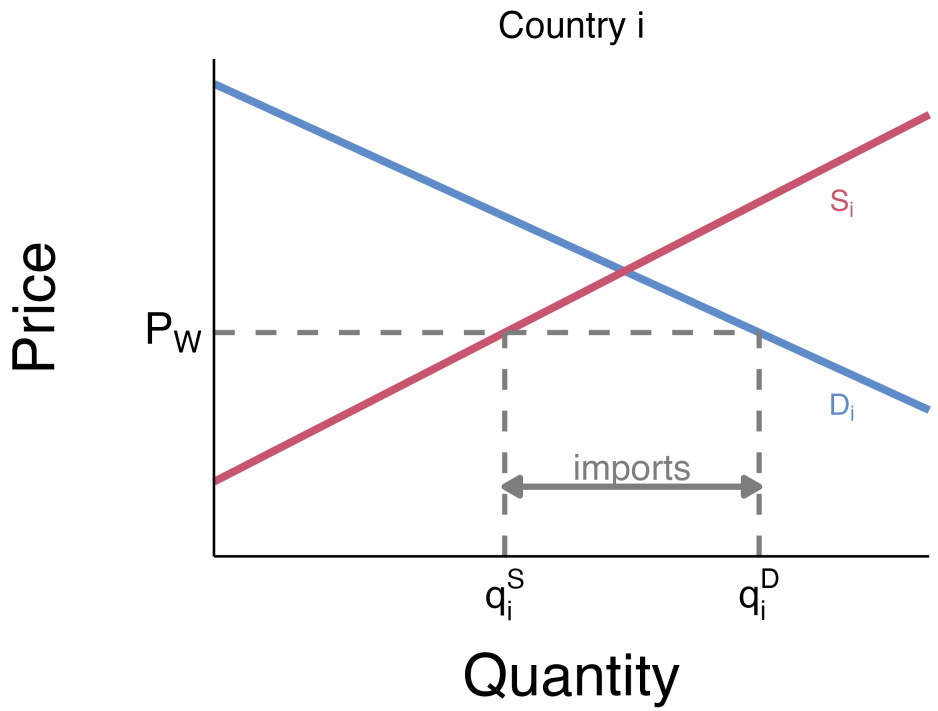
Suppose country i imposes a carbon tax $t = e \times d$ on domestic production

This shifts country i's supply curve up by t

But the **world price doesn't change** (small open economy)

So the carbon tax reduces the net price domestic producers receive from P_W to $P_W - t$

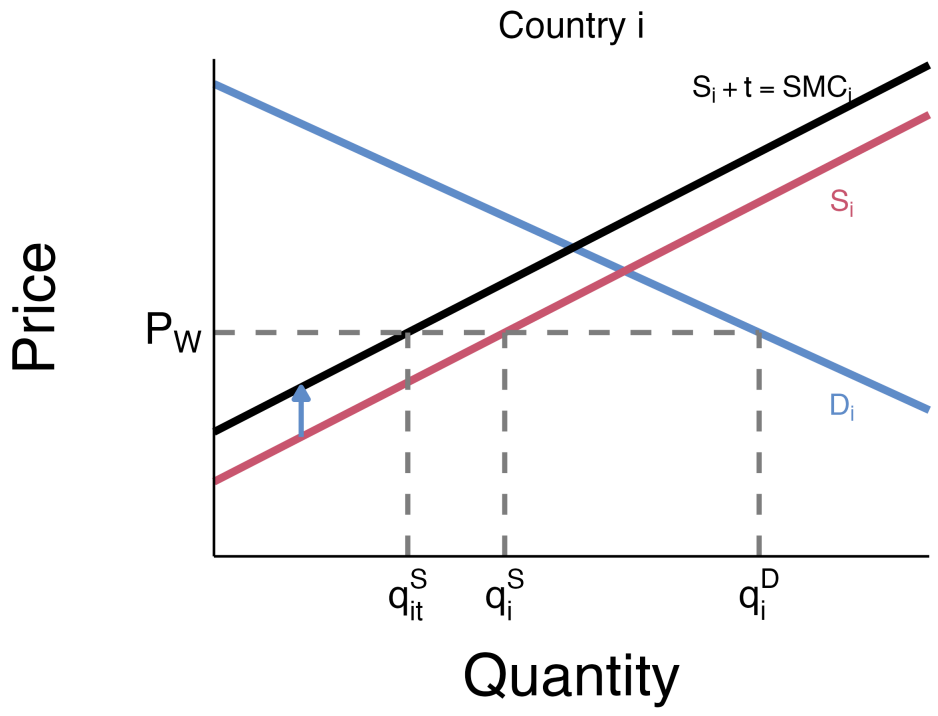
Effects of unilateral carbon tax



Before the tax, producers receive P_W and supply q_i^S

Their marginal production cost equals the price they receive

Effects of unilateral carbon tax



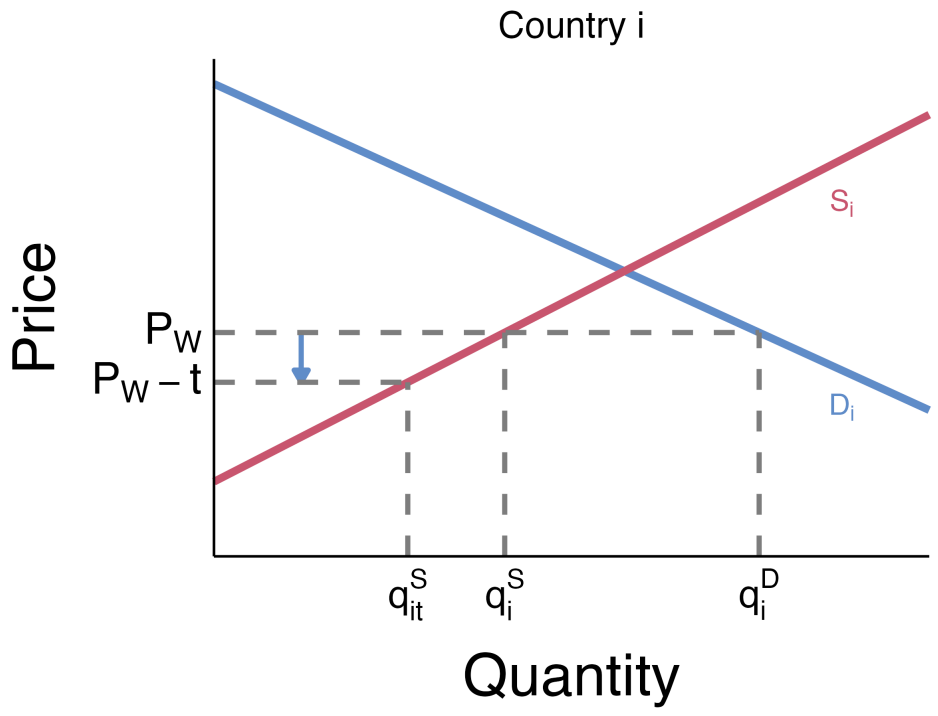
First view: shift supply up by the tax $t = e \times d$

Firms need a price that covers production cost plus tax

$$S_i(q) + t = P_W$$

Output falls to q_{it}^S

Effects of unilateral carbon tax



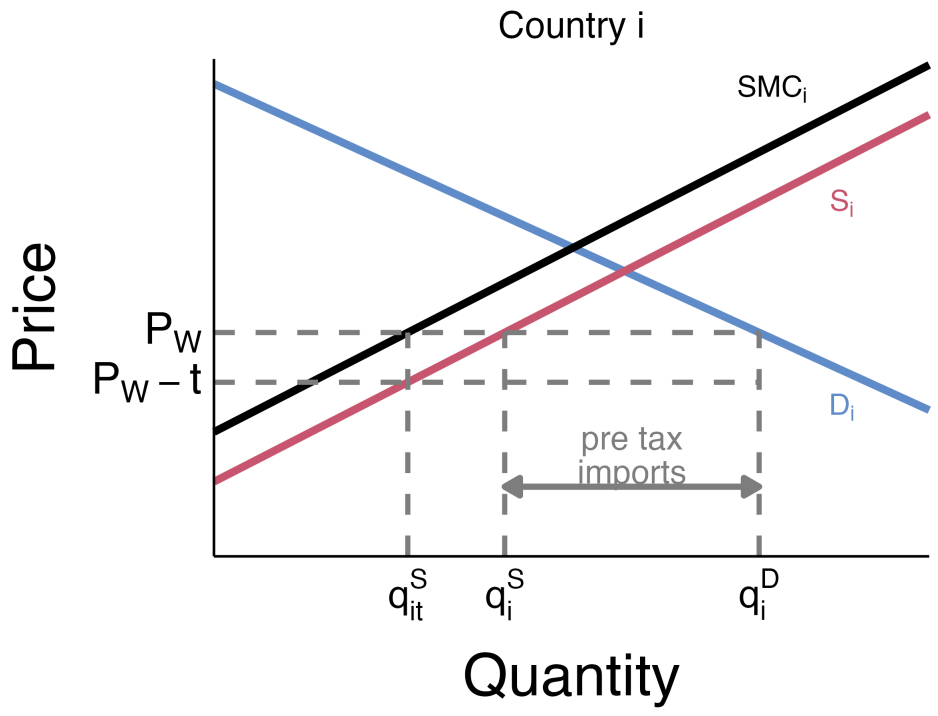
Equivalent view: keep the original supply curve

Producers receive P_W but pay t, leaving $P_W - t$

$$S_i(q) = P_W - t$$

The same output q_{it}^S results

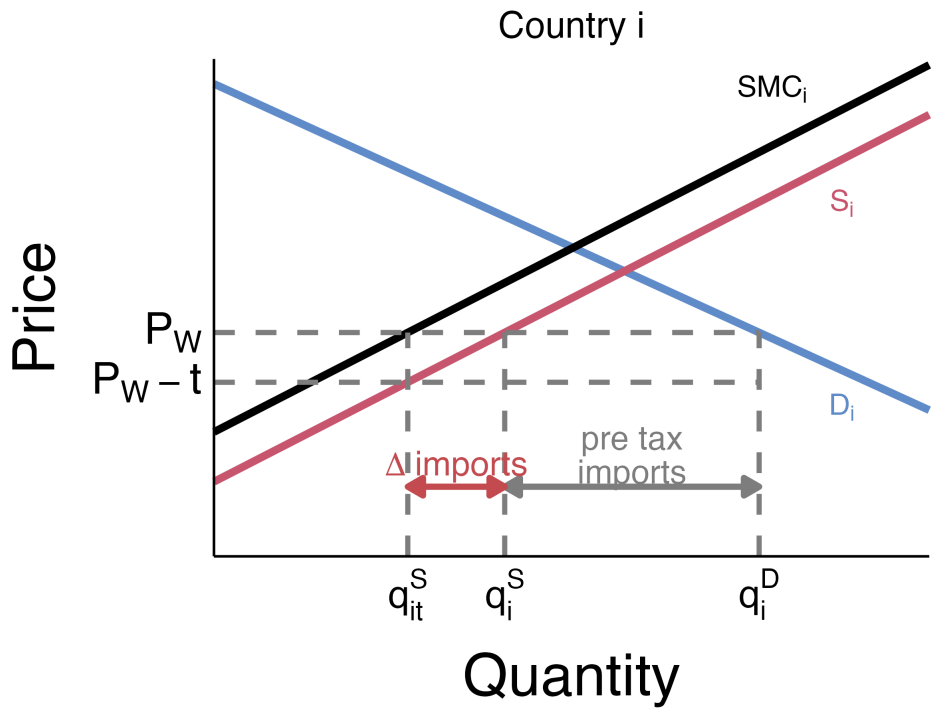
Effects of unilateral carbon tax



After the tax:

- Consumers still face the world price P_W
- But with a tax t, producers receive the net of tax price $P_W - t$
- Domestic output falls to q_{it}^S

Effects of unilateral carbon tax

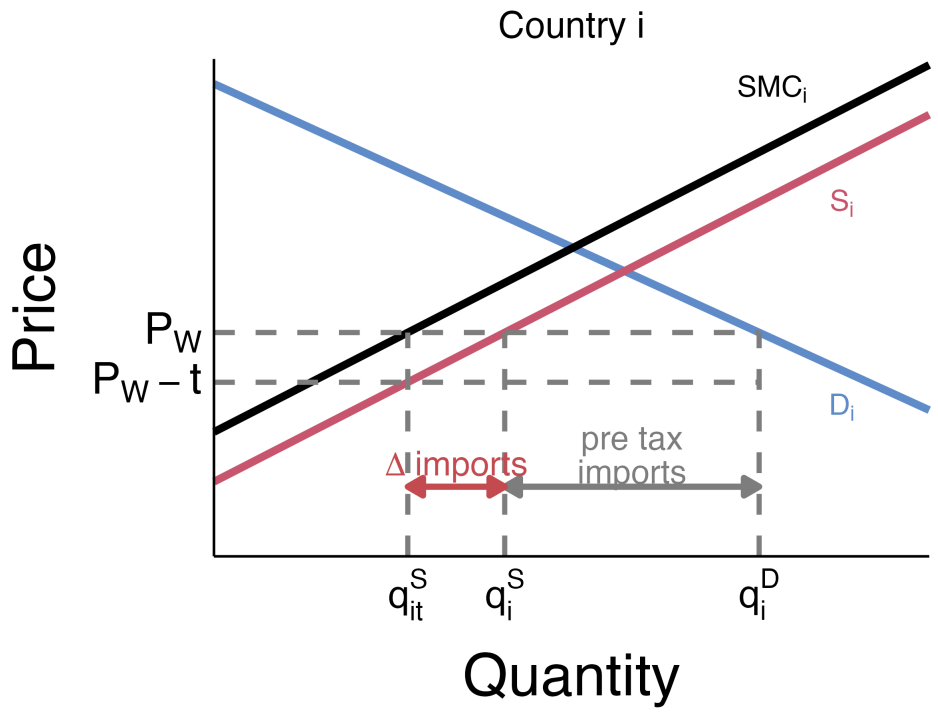


Leakage comes next:

- Consumers in country i still demand q_i^D at price P_W
- Imports rise enough to offset the fall in domestic output
- Foreign production replaces home production: that is leakage

So what happens to production and emissions globally?

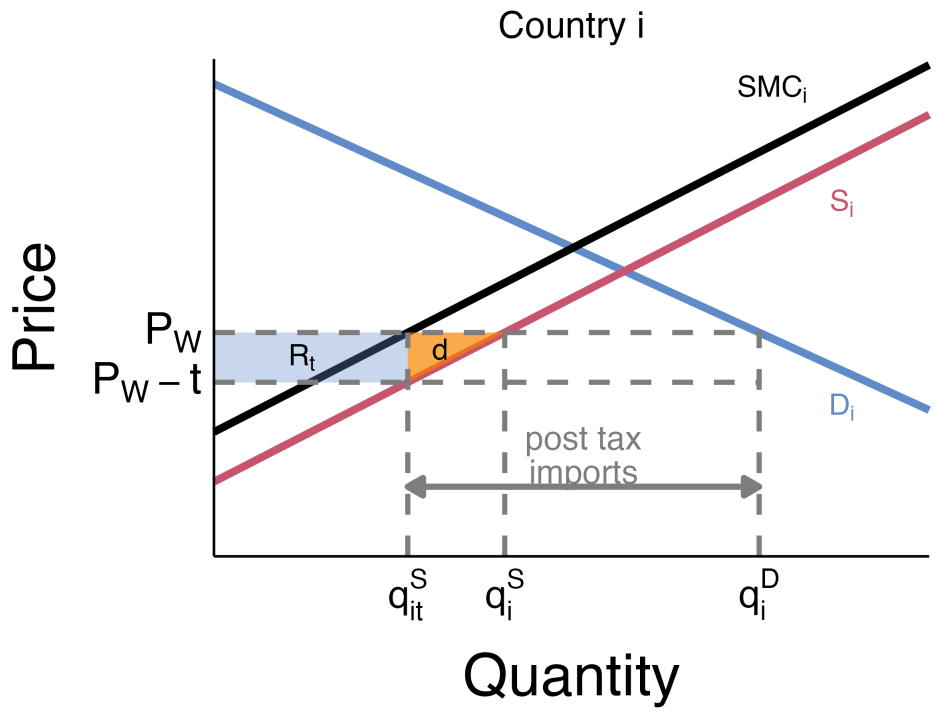
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Leakage comes next:

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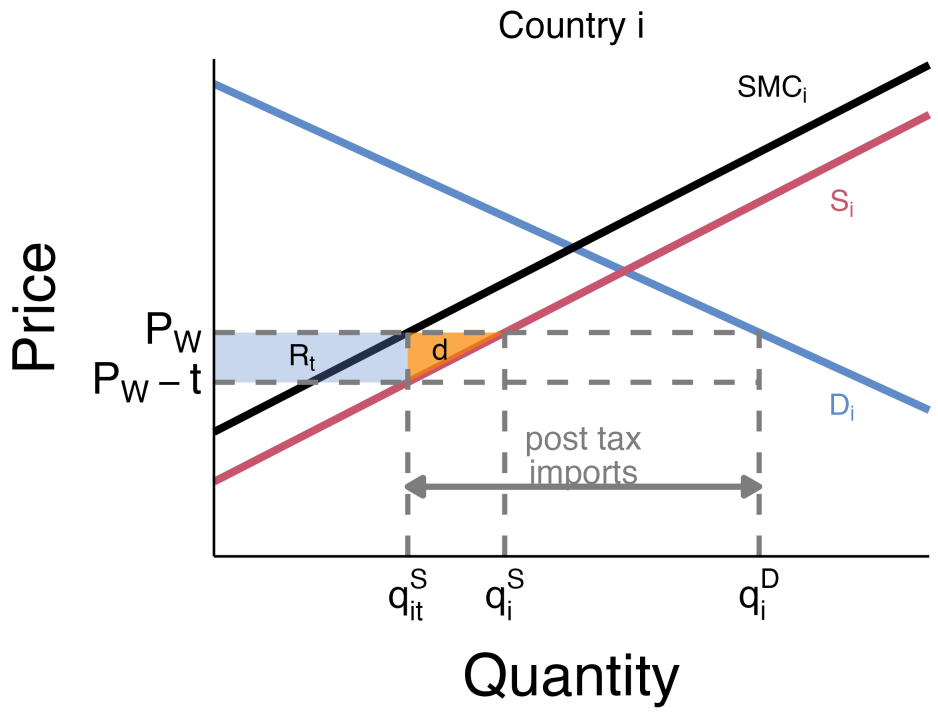
Effects of unilateral carbon tax



From country i's perspective:

- Consumer surplus does not change: they still pay P_W and consume q_i^D
- The government gains revenue R_t

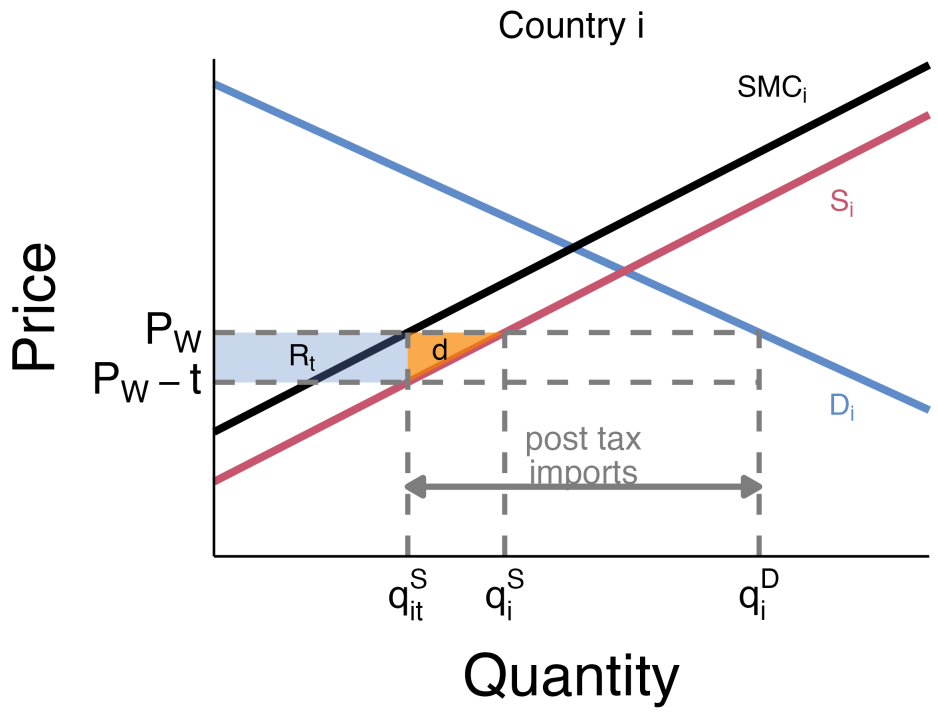
Effects of unilateral carbon tax



From country i's perspective:

- Producers lose producer surplus equal to $R_t + d$

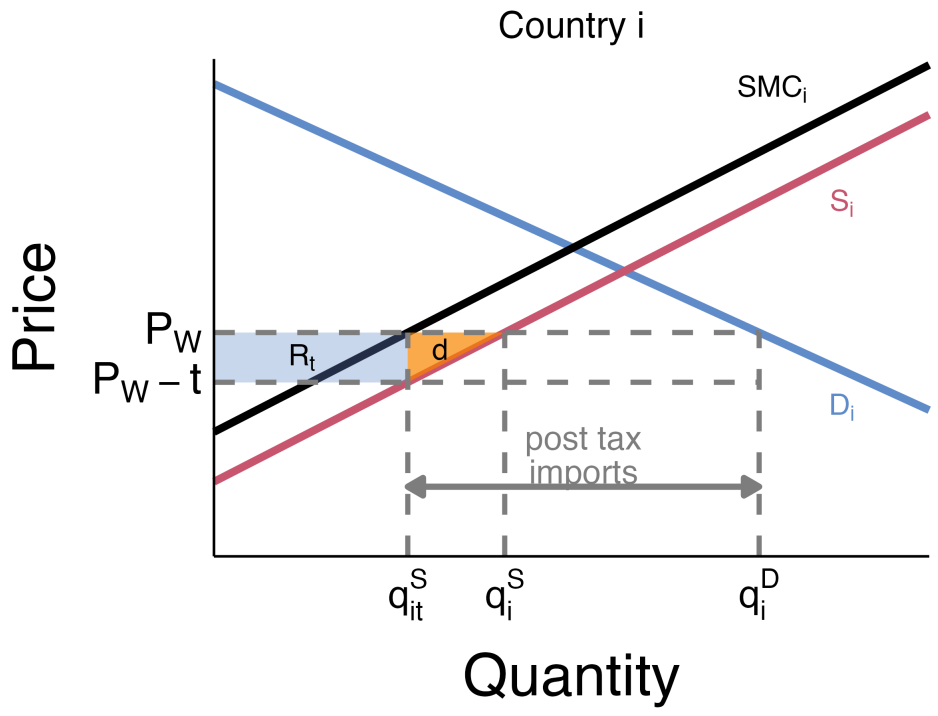
Effects of unilateral carbon tax



Why do producers lose $R_t + d$?

- Blue rectangle: they receive a lower net of tax price on the units they still sell
- Orange triangle: they also stop selling the units between q_{it}^S and q_i^S for which their PMC $< P_W$

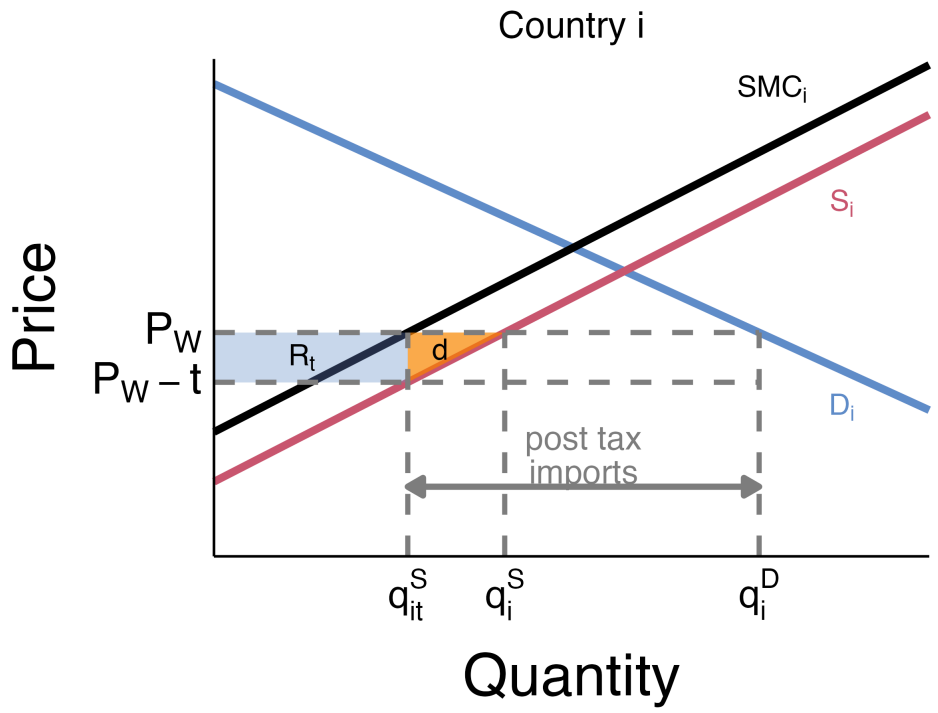
Effects of unilateral carbon tax



Why is d the country's deadweight loss?

- R_t is only a transfer from producers to the government, so it stays inside country i
- Net national effect: $+R_t - (R_t + d) = -d$

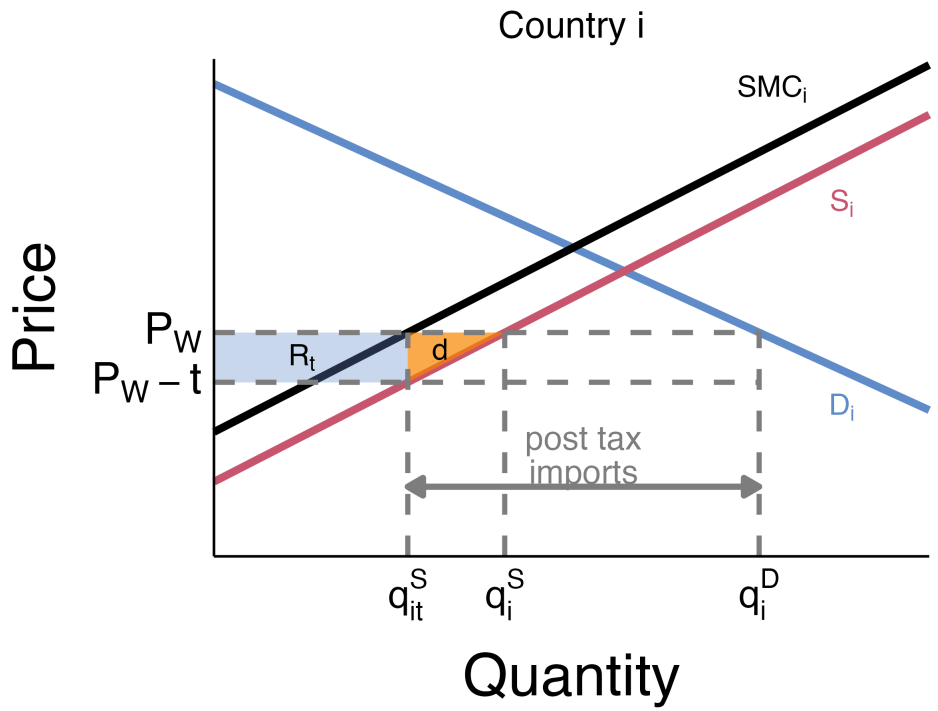
Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The triangle d is the pure private loss left after netting out the transfer R_t
- Consumers in country i are unaffected and imports replace the lost domestic output/emissions

Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The tax reduces home production, but foreign production rises 1-for-1: that is leakage
- The environmental gain is offset abroad, so country i is left with the private DWL d

Unilateral carbon tax effects

The increase in imports means **foreign production increases**

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- Production and emissions increase abroad

Leakage undermines the tax

With identical carbon intensities and a small country implementing the tax (so P_W is fixed):

- 1 unit fewer domestic production = 1 unit more foreign production
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Country i bears costs but gets no environmental gain

Leakage is a rate, not a constant

Economists often summarize leakage with the **leakage rate**:

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

where $\Delta E_i < 0$ is the change in home emissions and $\Delta E_j > 0$ is the change in foreign emissions

Leakage is a rate, not a constant

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

Benchmarks:

- 0: no foreign offset
- Between 0 and 1: global emissions still fall, but less than the domestic reduction
- 1: one for one offset, as in this small open economy example with identical intensities
- Greater than 1: possible if output shifts to dirtier foreign producers

What determines leakage?

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Leakage is **larger** when the world price moves less in response to the tax:

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- With little price movement, domestic demand doesn't change and imports replace the lost domestic output

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- Foreign supply is elastic and trade costs are low, so firms abroad can expand output quickly when home production falls
- Foreign production is more emissions intensive than home production ($e_j > e_i$), so each unit shifted abroad offsets more of the home emissions reduction

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Leakage is **smaller** when policy reduces quantity demanded by increasing the world price:

- The taxing country is large in world markets, so its supply cut changes the world price
- With price increases, domestic quantity demanded falls
- Then fewer imports are needed to fill the gap, so there is less of an emissions increase abroad, and greater global emissions reductions

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Leakage is **smaller** when foreign expansion is constrained or cleaner:

- If foreign producers supply is inelastic, firms abroad are limited in how much they can increase production
- If home production is dirtier than foreign production ($e_i > e_j$), shifting output abroad offsets less of the domestic abatement and can even lower global emissions

Carbon tax + carbon tariff

Adding a carbon tariff

The problem with the carbon tax was that consumers can easily switch to importing the dirty good from another country not subject to the carbon tax

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One potential way to address this is a **carbon tariff** of rate τ: a tax on imports proportional to their carbon content

Here we will tax imports at the same rate as domestic production:

$$\tau = t = e \times d$$

Adding a carbon tariff

Key intuition:

A carbon tax reduces the net price producers receive by t

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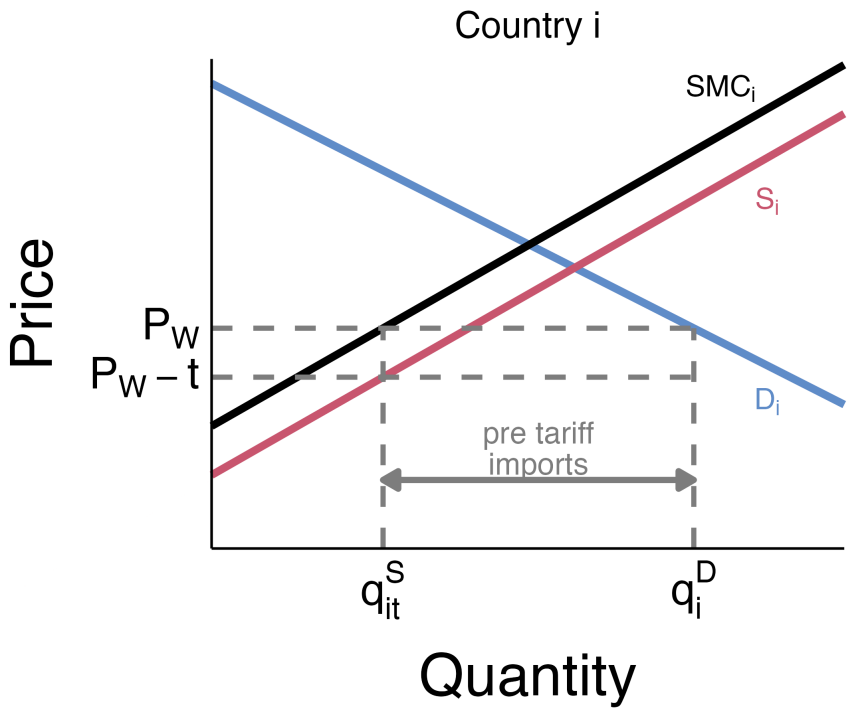
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So tax + tariff works like a consumption tax in the end

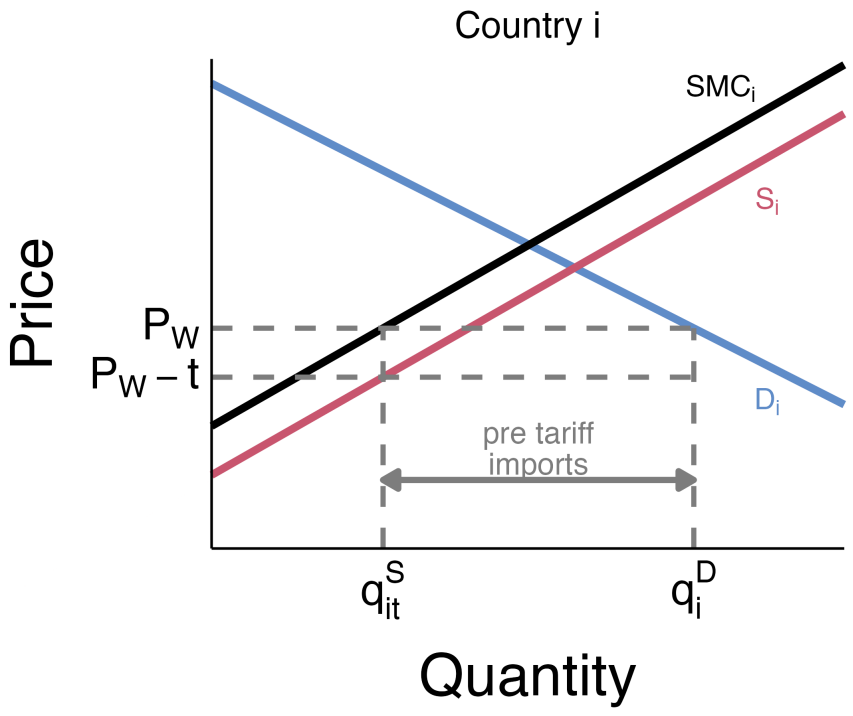
Adding a carbon tariff



Start from the carbon tax alone:

- Producers net $P_W - t$
- Home output is q_{it}^S
- Consumers still pay P_W and buy q_i^D
- Country i imports $q_i^D - q_{it}^S$

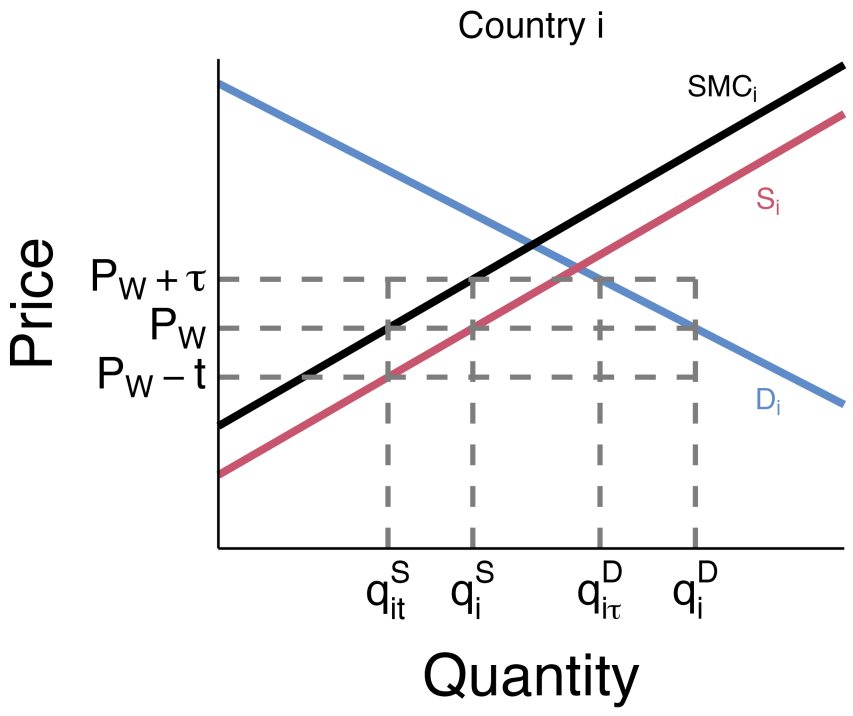
Adding a carbon tariff



Why is leakage a problem here?

- Imports replace the lost domestic output
- So home emissions fall, but foreign production rises
- The tax changes domestic supply, not consumption
- Since consumption doesn't change, there must be more imports

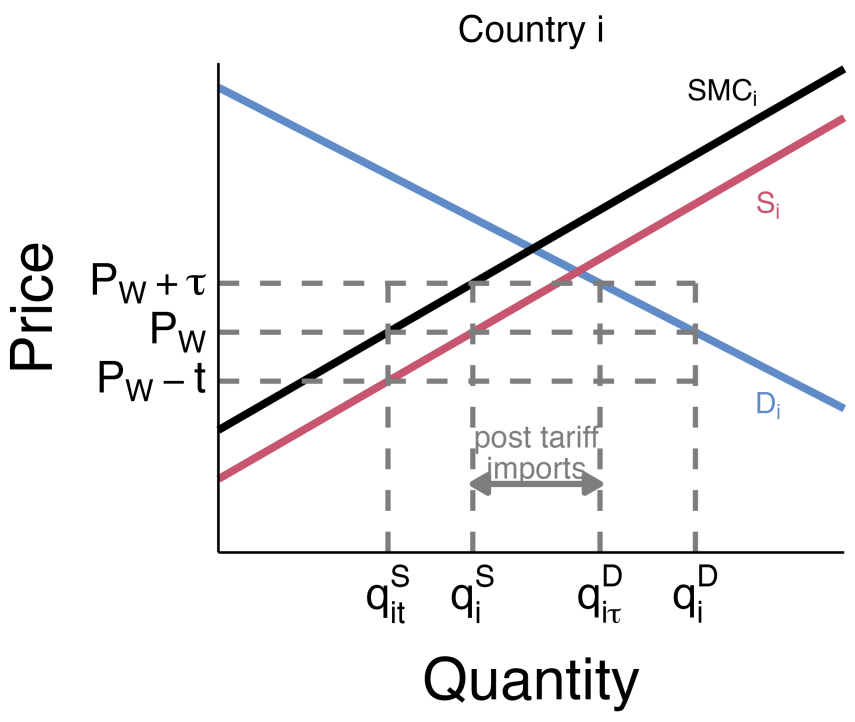
Adding a carbon tariff



Now add a carbon tariff $\tau = t$ on imports:

- Consumers pay $P_W + \tau$
- Domestic quantity demanded falls from q_i^D to $q_{i\tau}^D$
- Domestic producers receive $P_W + \tau - t = P_W$
- So domestic supply returns to q_i^S

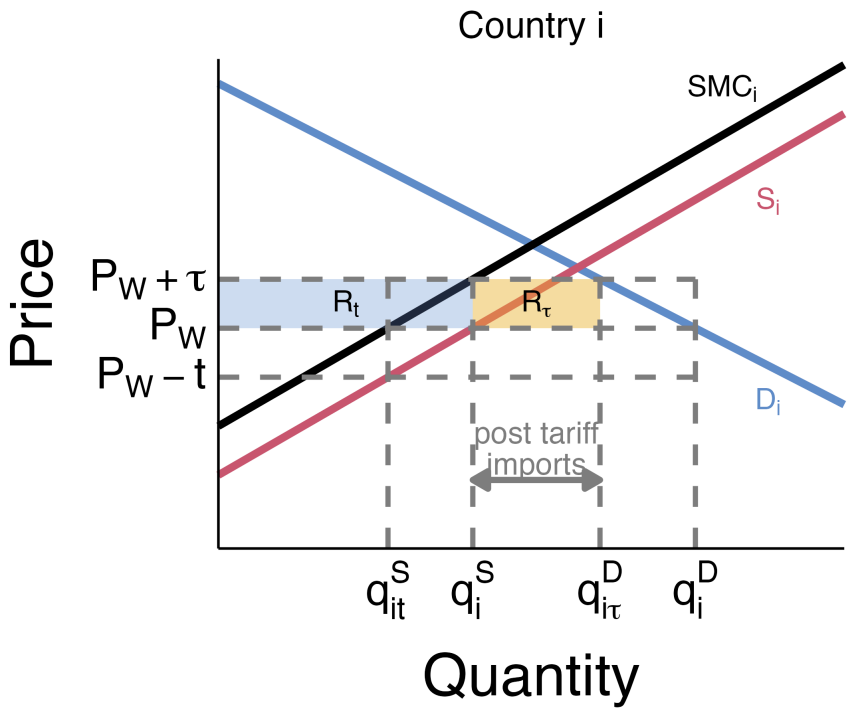
Adding a carbon tariff



Post tariff imports equal $q_{i\tau}^D - q_i^S$

- Higher domestic production and lower consumption reduce imports relative to the carbon tax alone
- Domestic production returns to its unregulated level, while consumption falls: **global production and emissions fall**

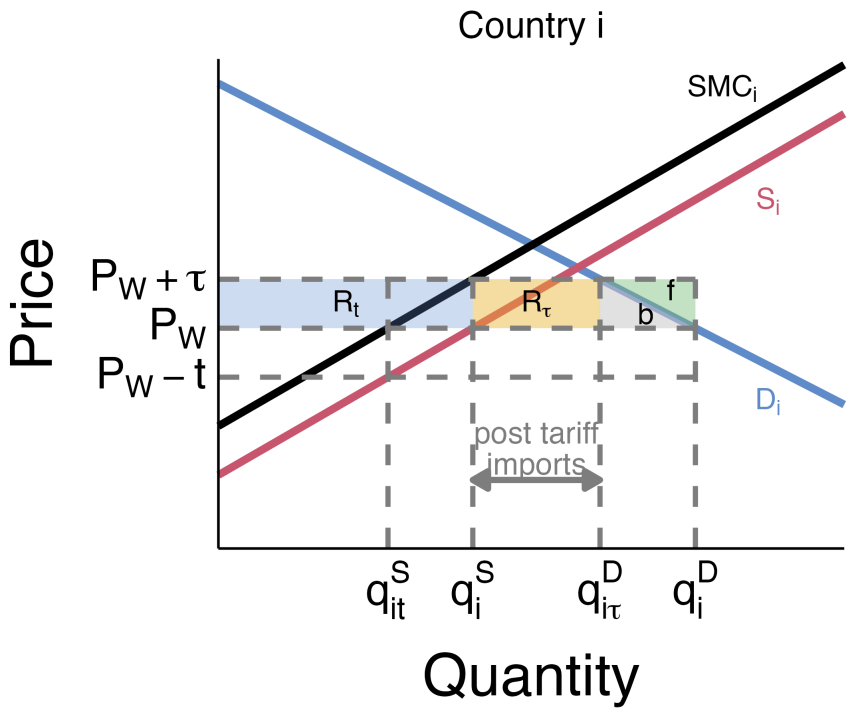
Adding a carbon tariff



Revenue effects:

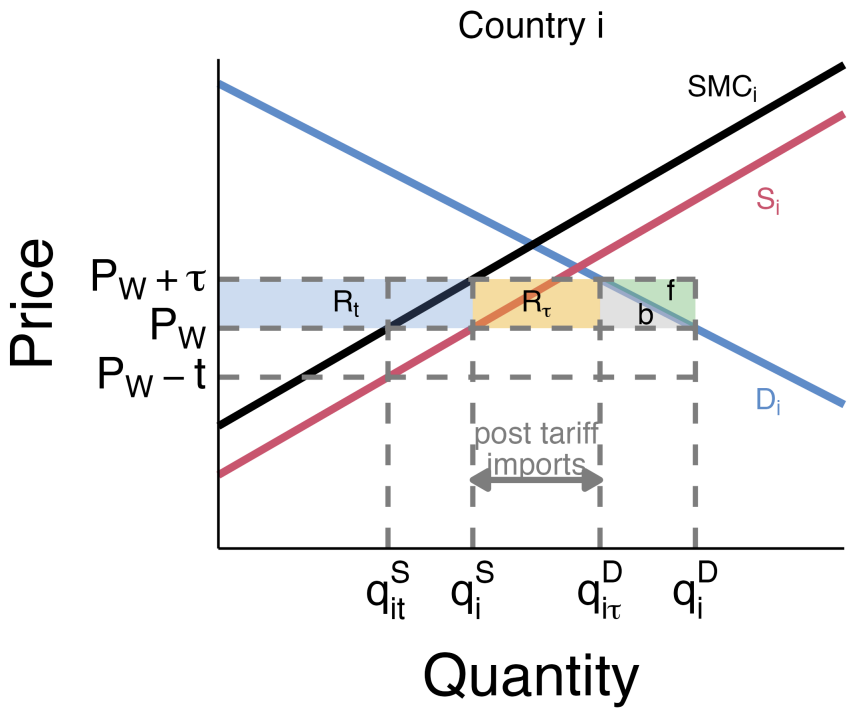
- The domestic carbon tax still raises R_t
- The tariff raises R_τ on remaining imports
- Both are transfers inside country i: tax R_t paid by producers, tariff R_τ paid by consumers

Adding a carbon tariff



- b is the consumption deadweight loss from the higher price
- Reduced consumption where the private benefit $D[i]$ was greater than the PMC of production P_W

Adding a carbon tariff



Since $\tau = t = e \times d$

- The price $P_W + \tau$ includes the external damage cost
- $b + f$ is the environmental gain from lower world output and emissions of $q_i^D - q_{i\tau}^D$
- Tax + tariff improves social welfare by f

Recap

With only a tax:

- Domestic production/emissions become more expensive
- Consumers can easily switch to imports: "leakage"
- We taxed production but received no environmental benefits

Adding a carbon tariff:

- We are now making foreign production/emissions more expensive for our domestic consumers
- Consumers can no longer as easily switch to imports
- Emissions fall because they are correctly priced (good!)
- Consumption falls (bad!)

The key insight

With identical carbon intensities everywhere:

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Carbon tax + carbon tariff = **Domestic consumption tax**

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Why?

- Tax on domestic production raises domestic costs
- Tariff on imports raises import costs
- Both domestic and imported goods consumed at home face the same carbon price
- Equivalent to taxing domestic consumption of the carbon intensive good

Numerical example: setup

Use the same simple market for both equivalence results:

$$Q_D = 100 - P, \qquad Q_S = P - 20, \qquad P_W = 40$$

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- Q_D = domestic quantity demanded
- Q_S = domestic quantity supplied
- $M = Q_D - Q_S$ = imports
- P_W = world price

Numerical example: setup

Carbon and policy terms:

- e_H and e_F = home and foreign emissions per unit
- d = damage per unit of emissions
- t = domestic carbon tax on home production
- τ = carbon tariff on imports

$$e_H = e_F = 1, \qquad d = 10, \qquad t = \tau = 10$$

Numerical example: setup

Start from the free trade benchmark:

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Start from the free trade benchmark:

Step 1: under free trade, the domestic price equals the world price

$$P = P_W = 40$$

Numerical example: setup

Start from the free trade benchmark:

Step 1: under free trade, the domestic price equals the world price

$$P = P_W = 40$$

Step 2: plug P_W into demand and supply

$$Q_D = 100 - 40 = 60$$

$$Q_S = 40 - 20 = 20$$

$$M = Q_D - Q_S = 60 - 20 = 40$$

Numerical example: tax + tariff

Under a domestic carbon tax and equal tariff, work in three steps:

Numerical example: tax + tariff

Under a domestic carbon tax and equal tariff, work in three steps:

Step 1: solve for prices

Here P_c is the price paid by consumers and P_p is the net price received by domestic producers

$$P_c = P_W + \tau = 40 + 10 = 50$$

$$P_p = P_c - t = 50 - 10 = 40$$

Numerical example: tax + tariff

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 40 - 20 = 20$$

$$M = Q_D - Q_S = 50 - 20 = 30$$

Numerical example: tax + tariff

Step 3: compute government revenue

Here R_t is domestic carbon tax revenue and R_τ is tariff revenue

$$R_t = t \times Q_S = 10 \times 20 = 200$$

$$R_\tau = \tau \times M = 10 \times 30 = 300$$

$$R = R_t + R_\tau = 200 + 300 = 500$$

Numerical example: equivalent consumption tax

Now replace tax + tariff with a pure consumption tax of 10:

Numerical example: equivalent consumption tax

Now replace tax + tariff with a pure consumption tax of 10:

Step 1: solve for prices

Consumers pay the tax, so P_c rises, but producers still receive the world price

$$P_c = P_W + 10 = 40 + 10 = 50$$

$$P_p = P_W = 40$$

Numerical example: equivalent consumption tax

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 40 - 20 = 20$$

$$M = Q_D - Q_S = 50 - 20 = 30$$

Numerical example: equivalent consumption tax

Step 3: compute government revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

Numerical example: equivalent consumption tax

Step 3: compute government revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

So tax + tariff and a consumption tax generate the same outcome:

- same consumer price: $P_c = 50$
- same producer price: $P_p = 40$
- same domestic demand: $Q_D = 50$
- same domestic supply: $Q_S = 20$
- same imports: $M = 30$
- same total revenue: 500

Carbon tariff alone

What if we only use a tariff?

Some proposals call for carbon tariffs **without** a domestic carbon tax

What if we only use a tariff?

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What does this accomplish?

What if we only use a tariff?

Key intuition:

A carbon tariff raises the price producers receive by τ, which is the same thing as a production subsidy of τ

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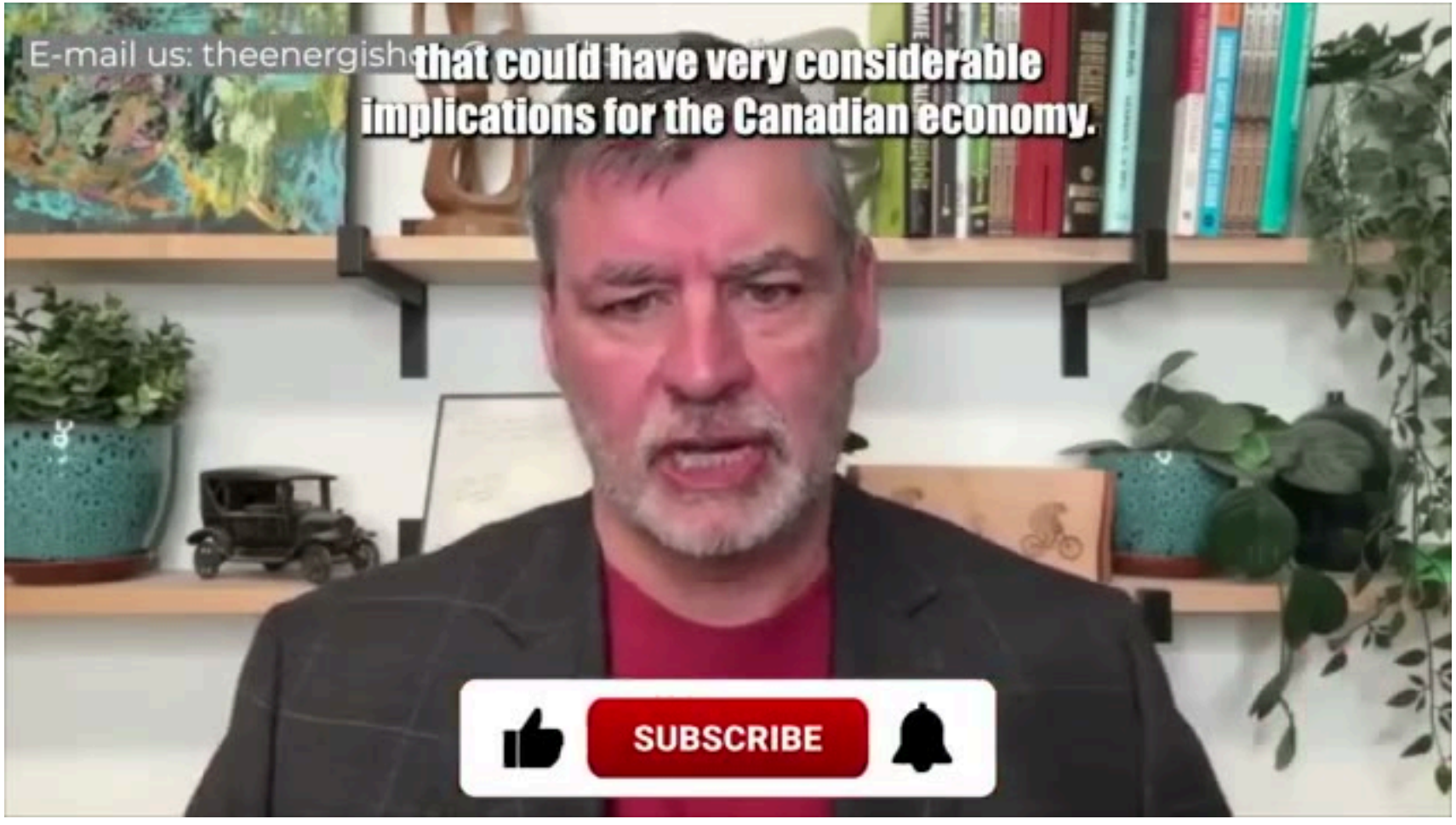
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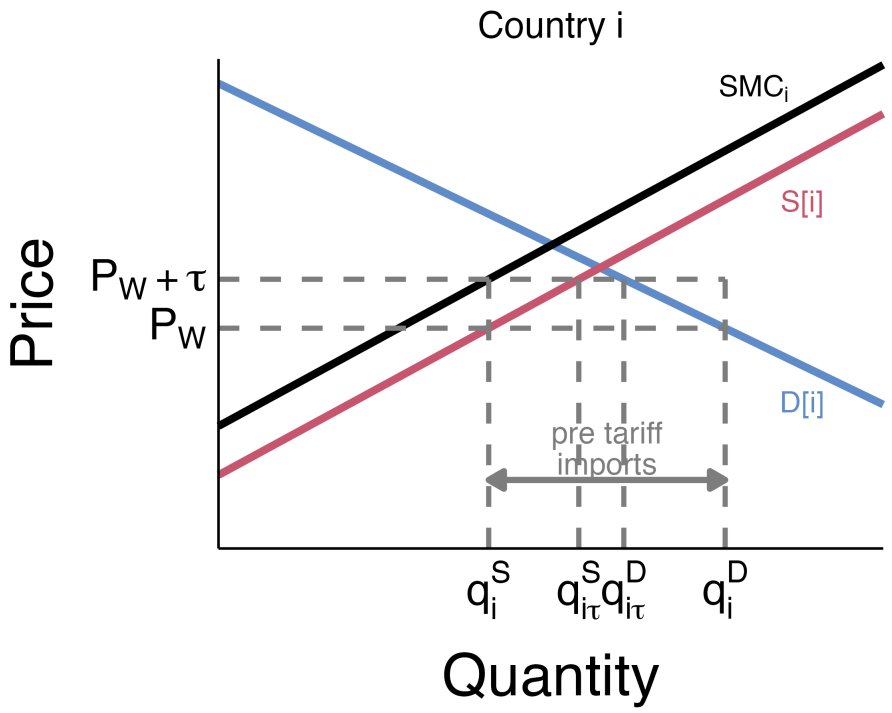
So tariff alone works like a consumption tax plus a production subsidy in the end

Carbon tariff alone in practice



Watch video

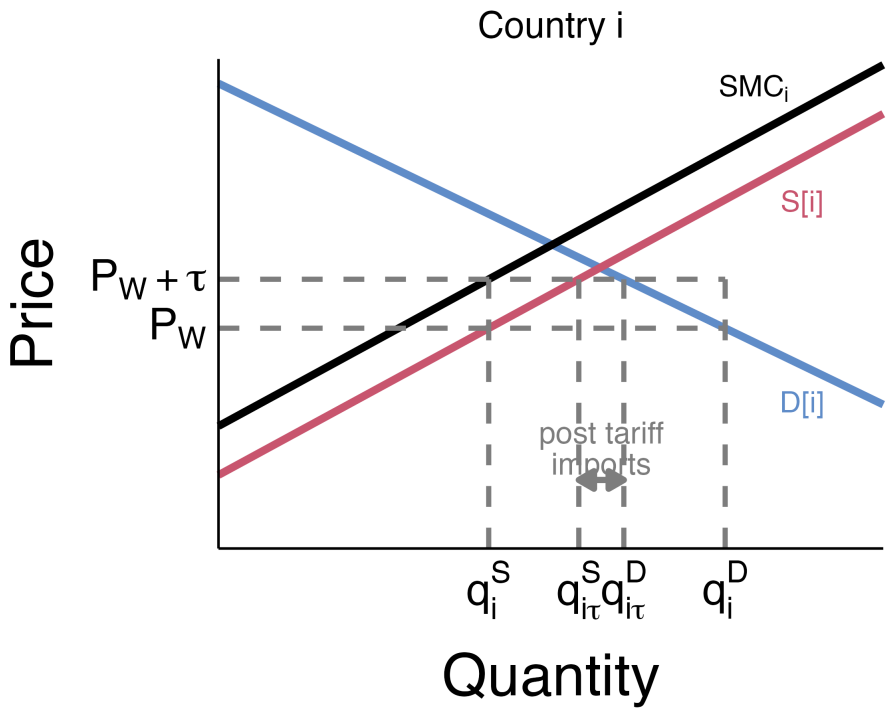
Carbon tariff alone



Start from free trade:

- Consumers and producers in country i both face P_W
- Domestic output is q_i^S and consumption is q_i^D
- Imports fill the gap $q_i^D - q_i^S$

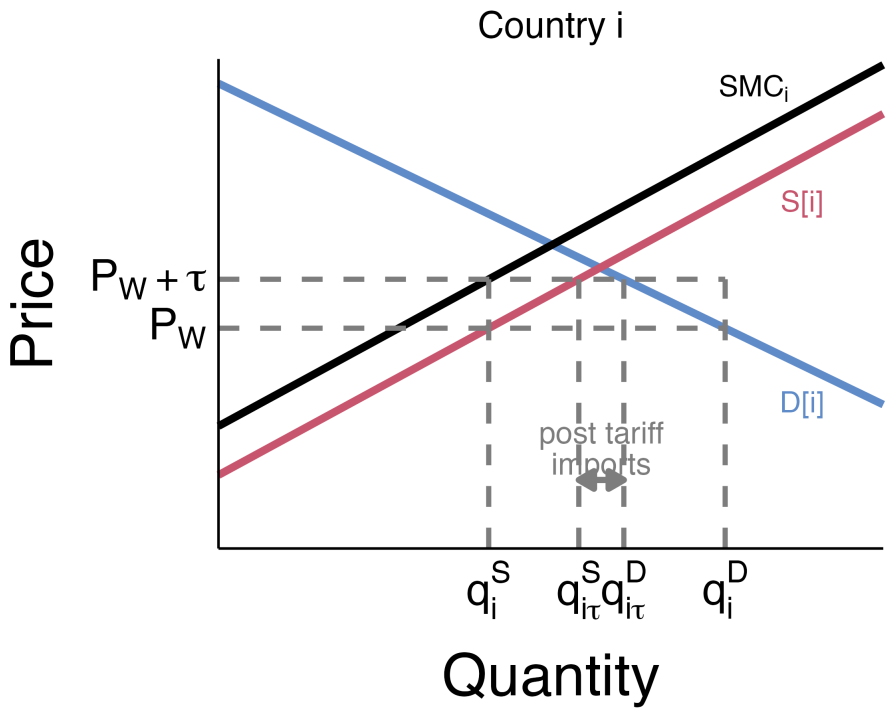
Carbon tariff alone



Now impose a carbon tariff τ with no domestic carbon tax:

- Consumers pay $P_W + \tau$
- Domestic producers also receive $P_W + \tau$
- So consumption falls and home output rises

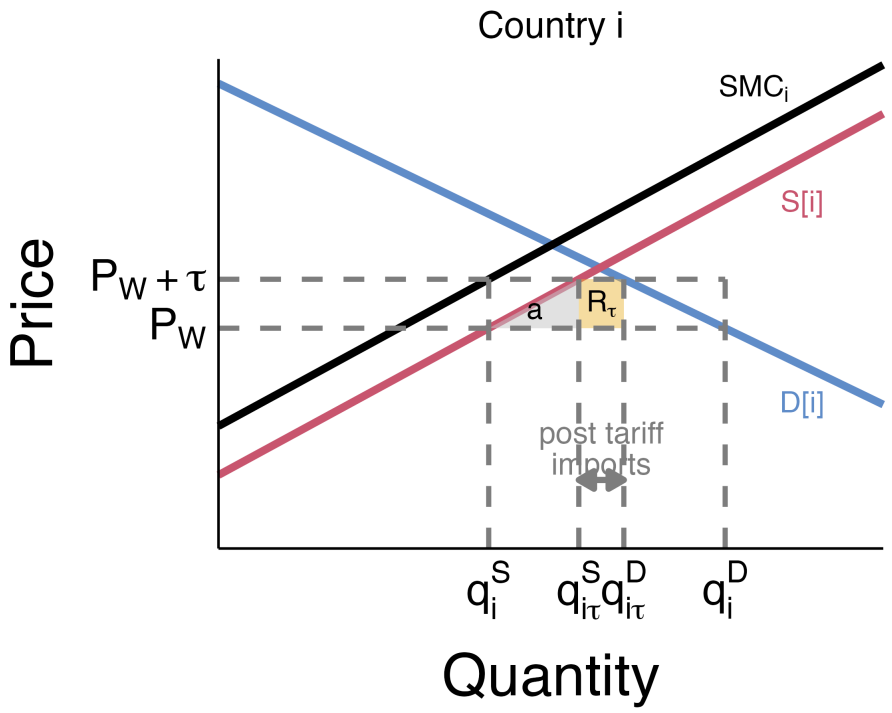
Carbon tariff alone



Why is this different from tax + tariff?

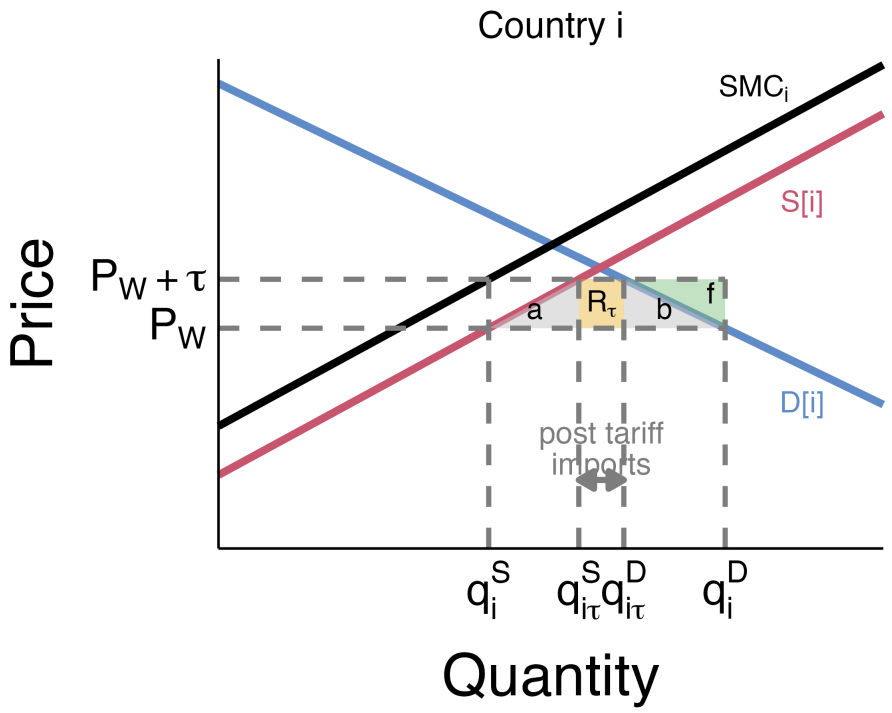
- The tariff alone taxes consumption but also protects domestic producers
- Output rises to $q_{i\tau}^S$ and demand falls to $q_{i\tau}^D$
- Imports shrink because both margins move

Carbon tariff alone



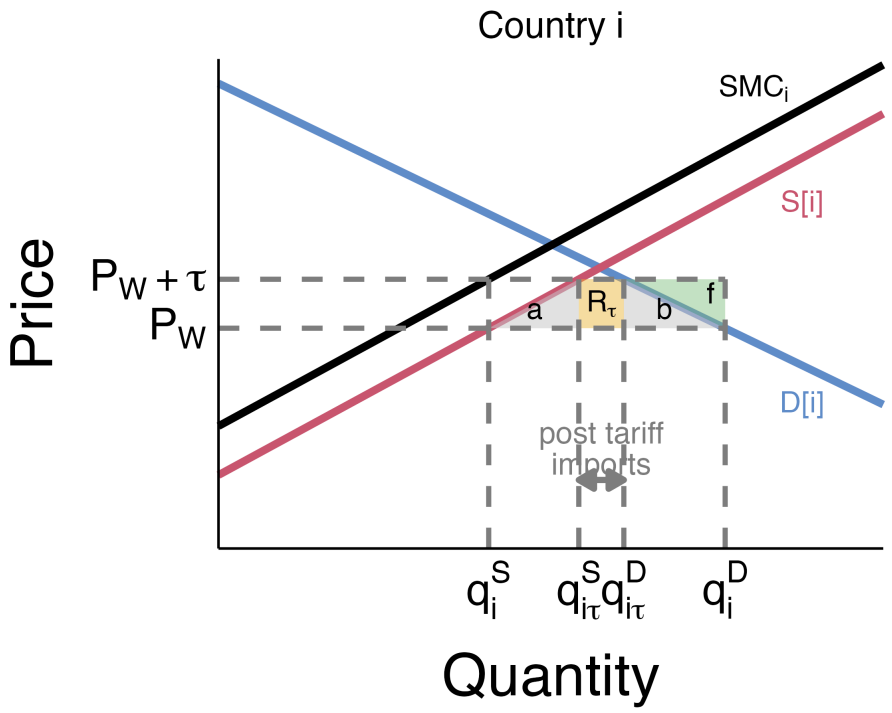
- The government collects tariff revenue R_τ on remaining imports
- The tariff pushes some production away from cheaper foreign suppliers

Carbon tariff alone



- a is the production DWL from replacing cheaper imports with costlier home output
- b is the consumption DWL from reduced consumption at the higher price
- Lower consumption has environmental benefits $b + f$: $(q_i^D - q_{i\tau}^D) \times d$

Carbon tariff alone



Welfare and environment:

- Net social benefit is $(b + f) - (b + a) = f - a$
- Gains from real reductions in pollution
- Losses from inefficient subsidization of domestic production
- Net effect is ambiguous

Numerical example: setup

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Numerical example: setup

For the tariff alone example, use the same market structure:

$$Q_D = 100 - P, \qquad Q_S = P - 20, \qquad P_W = 40$$

- Q_D = domestic quantity demanded
- Q_S = domestic quantity supplied
- $M = Q_D - Q_S$ = imports
- P_c = price paid by consumers
- P_p = price received by domestic producers

Policy for this case:

$$t = 0, \qquad \tau = 10$$

Numerical example: setup

Start from the free trade benchmark:

Numerical example: setup

Start from the free trade benchmark:

Free trade benchmark:

$$P_c = P_p = P_W = 40$$

$$Q_D = 100 - 40 = 60$$

$$Q_S = 40 - 20 = 20$$

$$M = Q_D - Q_S = 40$$

Numerical example: tariff alone

With a tariff alone, work in three steps:

Numerical example: tariff alone

With a tariff alone, work in three steps:

Step 1: solve for prices

Here both consumers and domestic producers face the tariff inclusive price

$$P_c = P_W + \tau = 40 + 10 = 50$$

$$P_p = P_c = 50$$

Numerical example: tariff alone

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 50 - 20 = 30$$

$$M = Q_D - Q_S = 50 - 30 = 20$$

Numerical example: tariff alone

Step 3: compute tariff revenue

$$R_\tau = \tau \times M = 10 \times 20 = 200$$

Numerical example: equivalent tax + subsidy

Now replace the tariff with a consumption tax of 10 and a production subsidy of 10:

Numerical example: equivalent tax + subsidy

Now replace the tariff with a consumption tax of 10 and a production subsidy of 10:

Step 1: solve for prices

Consumers pay the consumption tax, and producers receive the world price plus the subsidy

$$P_c = P_W + 10 = 40 + 10 = 50$$

$$P_p = P_W + 10 = 40 + 10 = 50$$

Numerical example: equivalent tax + subsidy

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 50 - 20 = 30$$

$$M = Q_D - Q_S = 50 - 30 = 20$$

Numerical example: equivalent tax + subsidy

Step 3: compute the government budget

Here R_c is consumption tax revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

$$\text{Subsidy cost} = 10 \times Q_S = 10 \times 30 = 300$$

$$R^{net} = R_c - \text{Subsidy cost} = 500 - 300 = 200$$

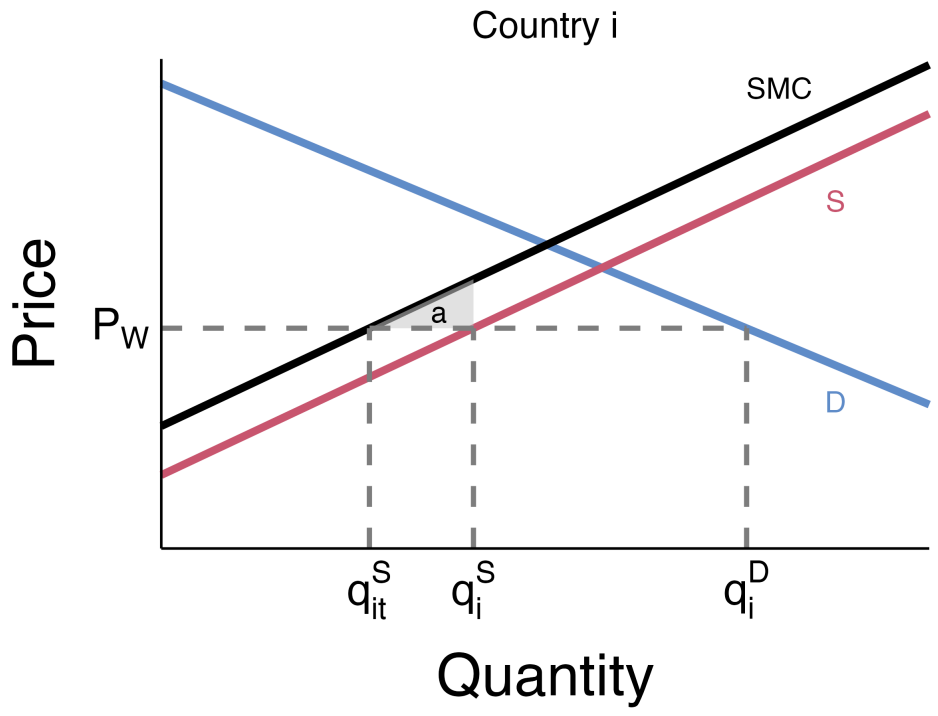
Numerical example: equivalent tax + subsidy

So tariff alone and a consumption tax plus production subsidy generate the same outcome:

- same consumer price: $P_c = 50$
- same producer price: $P_p = 50$
- same domestic demand: $Q_D = 50$
- same domestic supply: $Q_S = 30$
- same imports: $M = 20$
- same net government revenue: 200

World carbon tax

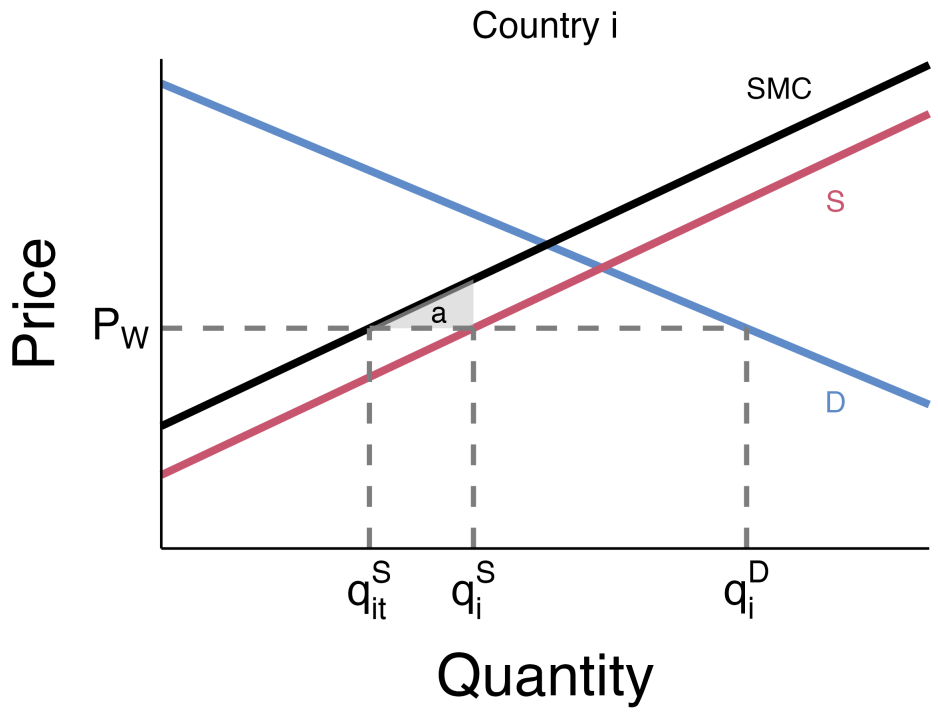
World carbon tax



Suppose the rest of the world already prices carbon so P_W reflects the carbon cost *abroad*:

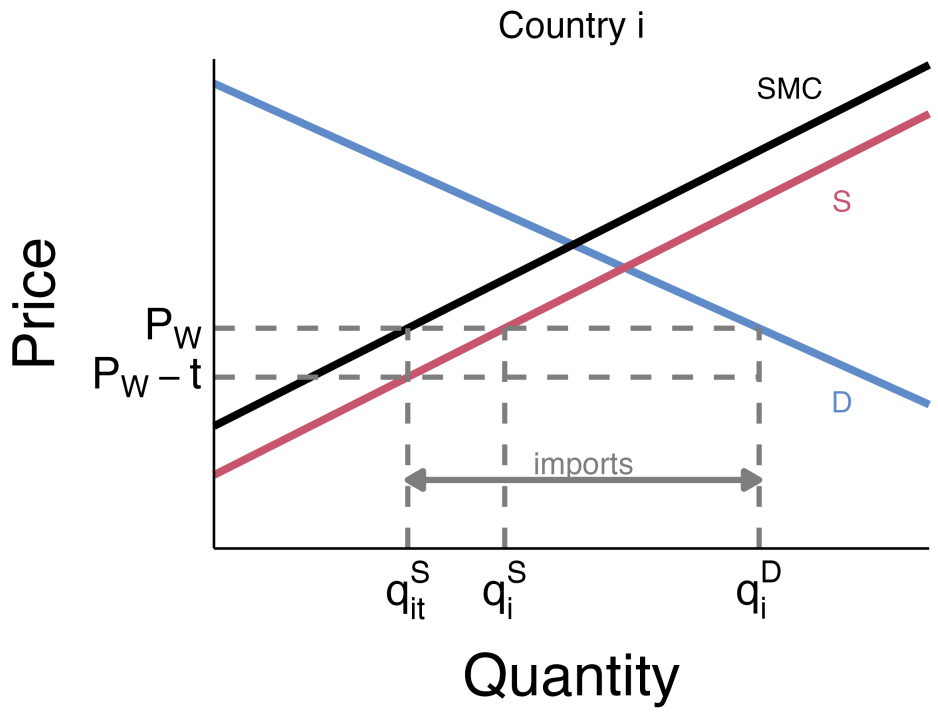
- Country i still produces q_i^S without its own carbon tax
- The efficient output for country i is q_{it}^S, where SMC = P_W

World carbon tax



- At q_i^S, SMC exceeds the revenue from selling the good at P_W
- $q_i^S - q_{it}^S$ is leakage *into* the country from carbon taxation abroad
- a is the resulting deadweight loss from overproduction domestically

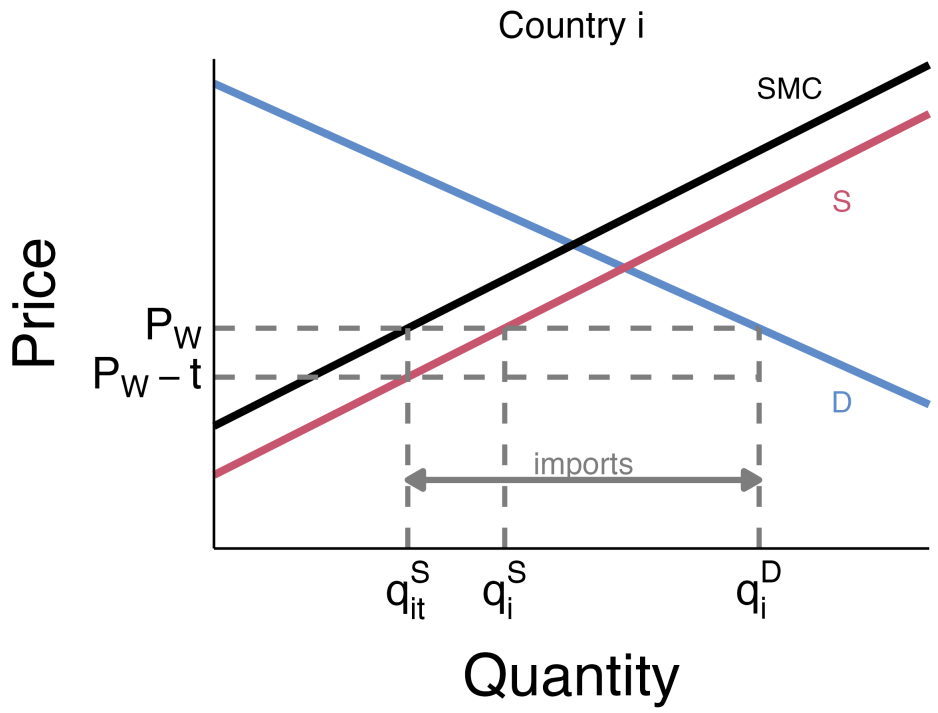
World carbon tax



If country i also imposes the carbon tax t:

- Producers receive net of tax price $P_W - t$
- Supply falls from q_i^S to q_{it}^S
- Imports rise to $q_i^D - q_{it}^S$
- Consumers still face the world price P_W

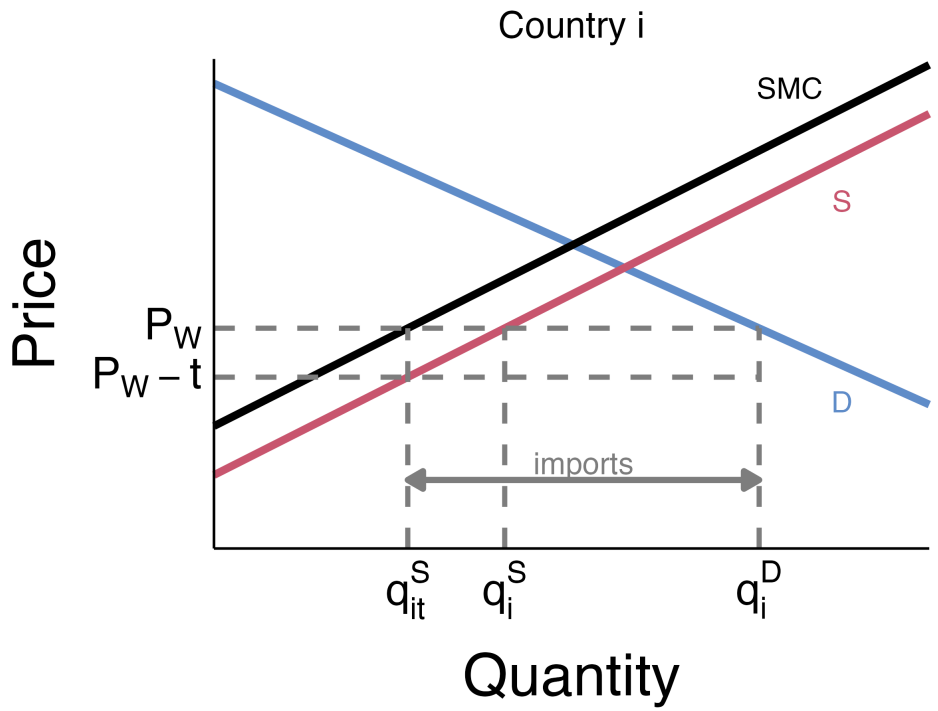
World carbon tax



Why are those extra imports not leakage?

- Foreign producers already face the same carbon price
- The earlier distortion was leakage from the rest of the world to country i
- Those imports reverse that leakage
- The DWL a disappears

World carbon tax



At that point the allocation is efficient:

- Every country has internalized the externality
- Global SMC equals the marginal benefit of the last unit consumed
- Production is cost effective across countries
- A carbon tariff is unnecessary

Complications

Real world complications

Several factors complicate carbon tariff design:

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4. **Retaliation:** trading partners may impose counter tariffs

Retaliation risk

Trading partners may retaliate



Resource shuffling

Suppose country j has both clean and dirty production

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With a carbon tariff based on average intensity:

- Export the clean stuff to country i
- Consume the dirty stuff domestically

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This is **another form of leakage**

Non price policies

Many countries use regulations rather than carbon prices:

- Fuel economy standards
- Renewable portfolio standards
- Building codes

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How do you calculate the "equivalent carbon price"?

Non price policies

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- Fuel economy standards
- Renewable portfolio standards
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How do you calculate the "equivalent carbon price"?

Very difficult: may require complex modeling

Export rebates

Should country i rebate carbon taxes on exports?

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Arguments for:

- Maintains competitiveness
- Symmetric with import tariffs

Export rebates

Should country i rebate carbon taxes on exports?

Arguments for:

- Maintains competitiveness
- Symmetric with import tariffs

Arguments against:

- Reduces domestic incentive to decarbonize
- May look like an export subsidy (WTO issues)

WTO compatibility

Carbon tariffs face WTO scrutiny:

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EU CBAM designed carefully to fit within WTO rules

Key takeaways

Summary

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2. **Carbon tax + tariff** is equivalent to a consumption tax

    - Reduces consumption of carbon intensive goods
    - Can improve welfare if environmental benefit exceeds distortion costs

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1. **Carbon tariffs** aim to address leakage and competitiveness concerns

2. **Carbon tax + tariff** is equivalent to a consumption tax

    - Reduces consumption of carbon intensive goods
    - Can improve welfare if environmental benefit exceeds distortion costs

3. **Carbon tariff alone** = consumption tax + production subsidy

    - May increase domestic emissions
    - Less effective than tax + tariff

Summary (continued)

1. **World carbon tax** is first best
    - Carbon tariffs are second best when global coordination fails

Summary (continued)

1. **World carbon tax** is first best
    - Carbon tariffs are second best when global coordination fails
2. **Many complications** in practice
    - Resource shuffling, non price policies, export rebates, WTO rules

The bottom line

Carbon tariffs are a **complement** to carbon pricing, not a substitute

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They work best when:

- Combined with a domestic carbon tax
- Set based on actual carbon content
- Part of broader international coordination

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They work best when:

- Combined with a domestic carbon tax
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- Part of broader international coordination

Alone, they are an imperfect tool that may create new distortions